GGG-P-643A
July 17, 1968
SUPERSEDING
Int. Fed. Spec. GGG-P-00643(GSA-FSS)
October 7, 1964 and
Fed. Spec. GGG-P-781A (In Part) (See 6.2)
November 7, 1958

FEDERAL SPECIFICATION

PULLER KIT, MECHANICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers mechanical puller kits which are commonly used for pulling bushings, bearings, gears, wheel hubs and cylinder sleeves. Some of the kits are also used for installing bushings, bearings, etc.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Puller kits shall be of the following types, classes, and styles, as specified (see 6.1):

Type I. Bearing and bushing, removal.

Class 1. Light duty.
Class 2. Medium duty.
Class 3. Heavy duty.

Type II. Driver plug, bearing and bushing, installing and removing.

Class 1. Engaging threaded adapter.

Style A. Light duty.
Style B. Heavy duty.

Class 2. Engaging bushing with shaft.
Class 3. Engaging bushing with two screws.

Type III. Cylinder sleeve, installing and removing.

Type IV. Wheel hub removal.

Type V. Puller attachment; bearing and bushing installing and removing kit.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specification and standards, of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

GGG-P-781 - Puller, Mechanical, Puller Attachment, Mechanical, and Puller Set, Mechanical.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 151 - Metals, Test Methods.

FSC 5120

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlinded under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specificatoons and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424 - Hand Tools, Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specification and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal shall apply:

Governmental:

National Bureau of Standards (NBS) Handbook:

H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Material. The chemical composition of the materials and the heat treatment employed shall be such as to produce puller kits complying with the requirements specified hereinafter for each type, class, style, and size. All materials shall be of uniform quality, free from all defects and imperfections which may affect their serviceability and durability.

3.2 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of puller kits which are otherwise in accordance with this specification. Dimensions shown on the illustrations are in inches.

3.3 Screw threads. All screw threads specified herein shall be in accordance with H-28.

3.4 Nuts and cap screws. Nuts and cap screws, when used, shall be of good quality steel and shall be assembled in a manner that excessive looseness and play will not develop under hard usage. The cap screws shall be of sufficient strength to resist breakage and deformation under the test load of the applicable table when the pullers are tested, as specified in section 4, to the applicable requirements, as specified in section 3.

3.5 Interchangeable parts and tolerances. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances, in order that such parts may be completely interchangeable, and may be replaced or adjusted without requiring modification. When no tolerances are specified herein, a tolerance of + 1/16 inch will be allowed for dimensions one inch and under and 1/8 inch for dimensions over one inch, provided the utility and interchangeability of the puller and its parts are not affected.

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3.6 Marking for identification Pullers shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

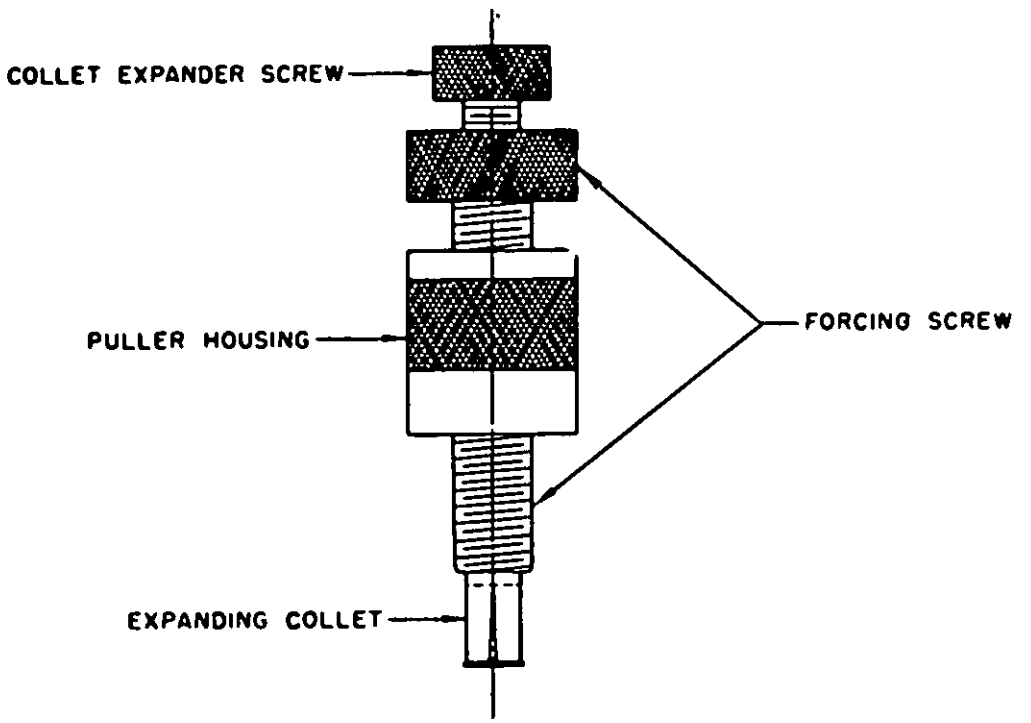
3.7 Carrying case. Carrying cases shall be provided with an identification, permanently and legibly marked with the following information on the outside of the cover:

Name of article.
Type, class, style.
Federal Specification number.
Manufacturer's kit number.
Manufacturer's name or trademark.
Property: United States Government.

3.8 Test loads. When test loads are specified, pullers shall withstand the minimum test load without failure of any part or parts of the puller. Failure shall be interpreted to be either breakage or deformation to such an extent that the puller loses its hold or grip or is otherwise rendered incapable of maintaining the load specified herein. When test loads are not specified, load testing is not required.

3.9 Type I, bearing and bushing, removal. Type I puller kits shall be suitable for removal of bearings and bushings from blind holes by use of expanding collets.

3.9.1 Class 1, light duty. Class 1 puller kits shall be capable of removing bearings and bushings having an inside diameter of not less than 1/8 nor more than 1/4 inch and shall consist essentially of a forcing screw, housing, collet expanding screw, and six expanding collets. The parts shall be of high-grade steel properly heat treated to resist breakage or deformation. Class 1 puller kit shall be similar to figure 1.



BEARING PULLER ASSEMBLY

FIGURE 1. Type I, class 1, puller kit, mechanical, bearing, and bushing, removal, light duty.

3.9.1.1 Forcing screw. The forcing screw shall conform to the dimensions shown on figure 2.

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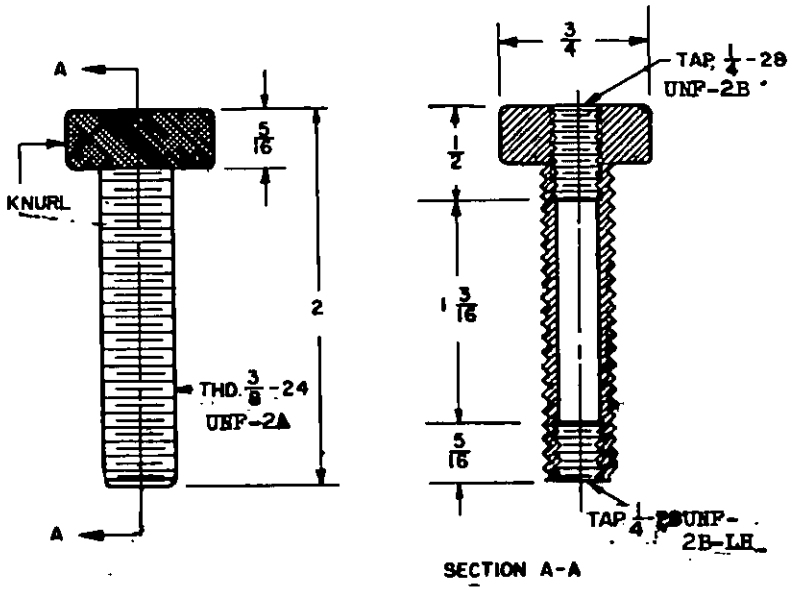


FIGURE 2. Forcing screw for type I, class 1 puller kit, mechanical, bearing and bushing, removal, light duty.

3.9.1.2 Housing. The housing shall conform to the dimensions shown on figure 3.

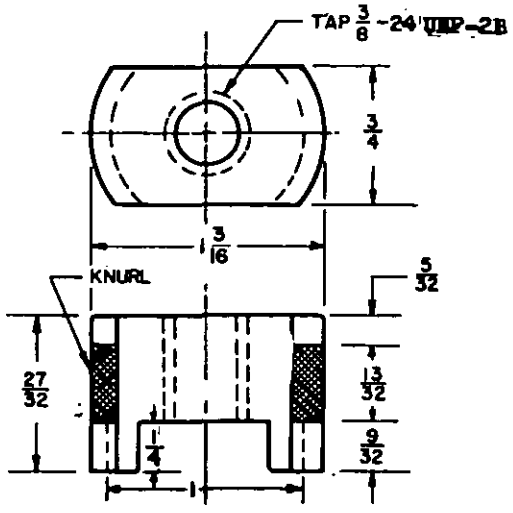


FIGURE 3. Housing for type I, class 1, puller kit, mechanical, bearing and bushing, removal, light duty.

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3.9.1.3 Collet expanding screw. The point of the screw shall be suitably shaped to expand the collets and the screw shall conform to the dimensions shown on figure 4.

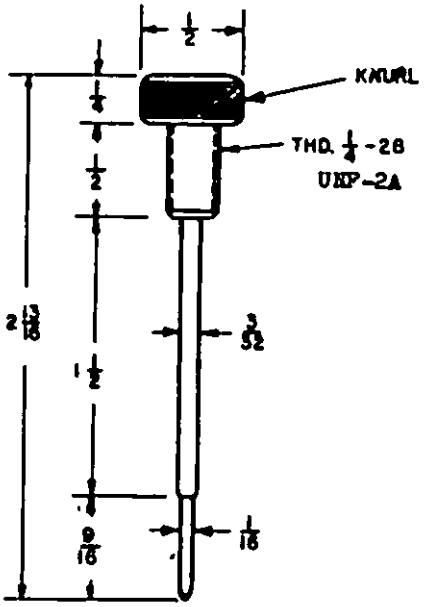
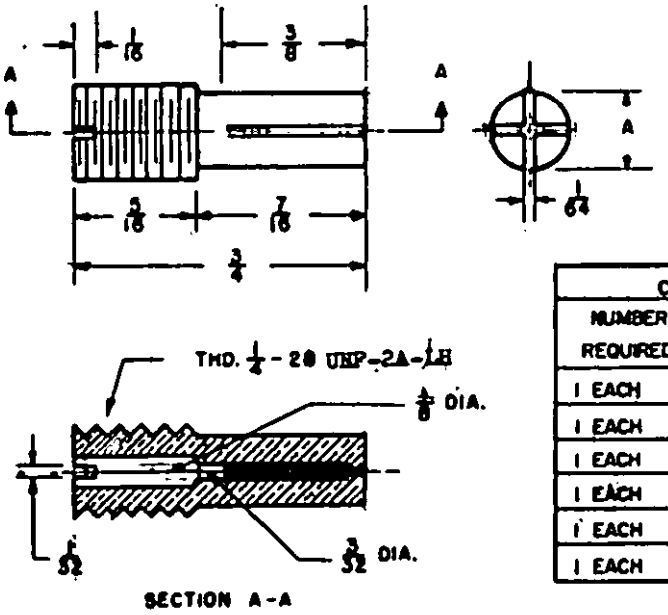


FIGURE 4. Collet expanding screw for type I, class 1, puller kit, mechanical, bearing and bushing, removal, light duty.

3.9.1.4 Collets. The center hole shall be properly formed to accept the expander screw. The collets shall conform to the dimensions shown on figure 5.



COLLETS	
NUMBER REQUIRED	DIMENSION A
1 EACH	1/8 INCH
1 EACH	4 MM
1 EACH	3/16 INCH
1 EACH	5 MM
1 EACH	6 MM
1 EACH	1/4 INCH

FIGURE 5. Collets for type I, puller kit, mechanical, bearing and bushing, removal, light duty.

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3.9.1.5 Carrying case. A carrying case shall be provided to accomodate the tools with individual holding spaces for the expanding collets. An instruction sheet, containing clear and concise instructions for proper operation of the tool, shall be attached to the side of each carrying case.

3.9.2 Class 2, medium duty. Class 2 puller kits, shall be capable of removing bearings and bushings having inside diameters of not less than 1/4 nor more than 1/2 inch and shall consist essentially of a draw nut, thrust washer, draw table, six expansion pins, adjusting screw, and six expanding collets. Class 2 puller kits shall be similar to figure 6.

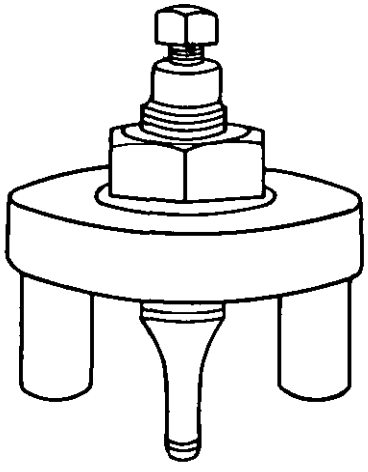


FIGURE 6. Type I, class 2, puller kit, mechanical bearing and bushing removal, medium duty.

3.9.2.1 Draw nut. The draw nut shall be of good-quality steel and shall be either casehardened to a depth of not less than 0.015-inch or heat treated to show not less than 40 on a Rockwell C scale. The draw nut shall be similar to figure 7, and shall conform to size 1 of table I.

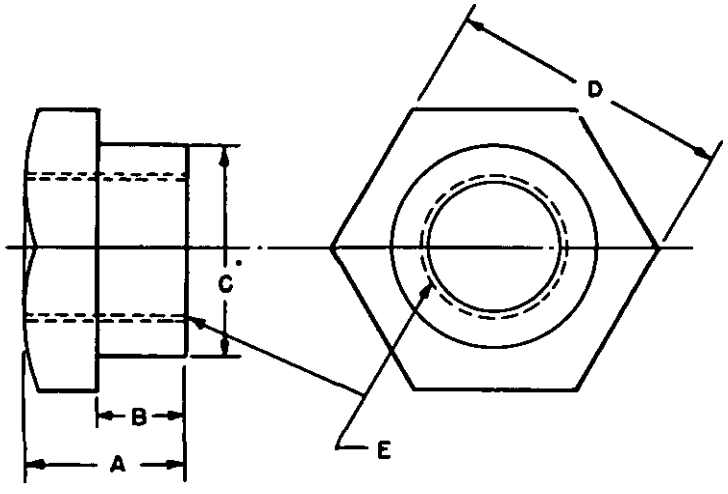


FIGURE 7. Draw nut for type I, classes 2 and 3, puller kit, mechanical, bearing and bushing removal, medium and heavy duty.

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TABLE I. Draw nut for type I, classes 2 and 3, puller kit, mechanical, bearing and bushing removal, medium and heavy duty

Size	Dimensions					
	A	B	C		D	E Thread UNF-2B
	Inches	Inch	Minimum	Maximum	Inches	Inches
1	1	1/2	0.993	0.998	1-1/4	3/4-16
2	1-1/8	5/8	1.493	1.498	2	1-1/4-12

3.9.2.2 Thrust washer. The thrust washer shall be of good fiber material, shall be similar to figure 8, and shall conform to size 1 of table II.

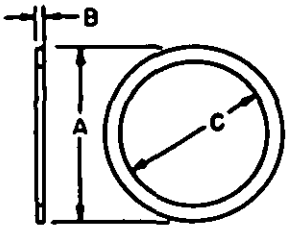


FIGURE 8. Thrust washer for type I, classes 2 and 3, puller kit, mechanical bearing and bushing, removal, medium and heavy duty.

TABLE II. Thrust washer for type I, classes 2 and 3 puller kit, bearing and bushing, removal, mechanical, medium and heavy duty

Size	Dimensions			
	A	B	C	
			Minimum	Maximum
	Inches	Inch	Inches	Inches
1	1-1/2	1/32	0.999	1.002
2	2	1/32	1.499	1.502

3.9.2.3 Draw table. The draw table for the class 2 puller kit shall be of cast iron having a table and three integral legs. The top surface and bottom of the legs shall be smoothly machined, parallel with each other within 0.010 inch. The draw table dimensions shall conform to figure 9.

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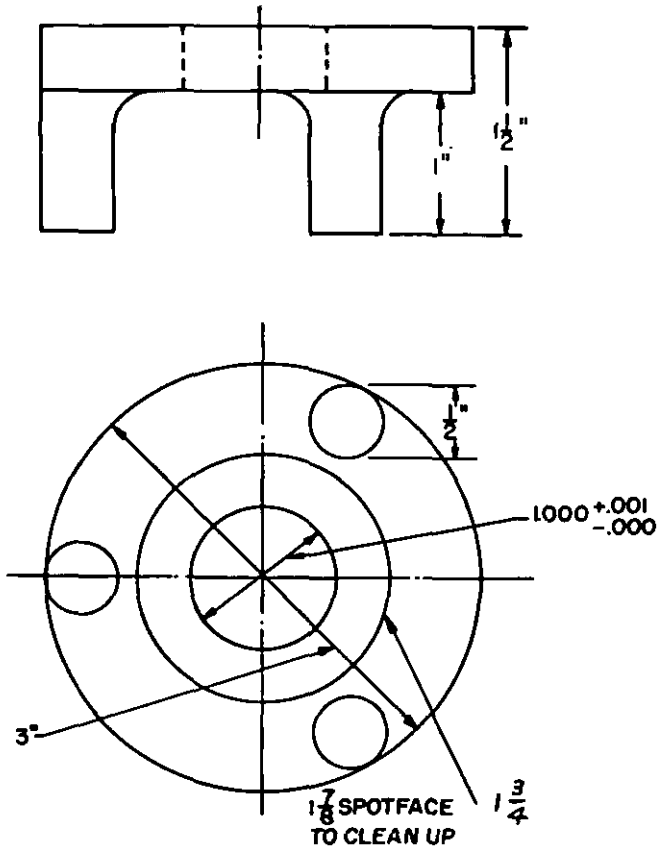


FIGURE 9. Draw table for type I, class 2, puller kit, mechanical, bearing and bushing, removal, medium duty.

3.9.2.4 Expansion pins. The expansion pins shall be of good quality steel and shall be heat treated to show not less than 50 on a Rockwell C scale on the spherical radius only. The expansion pins shall be similar to figure 10, and shall conform to size 1 of table III.

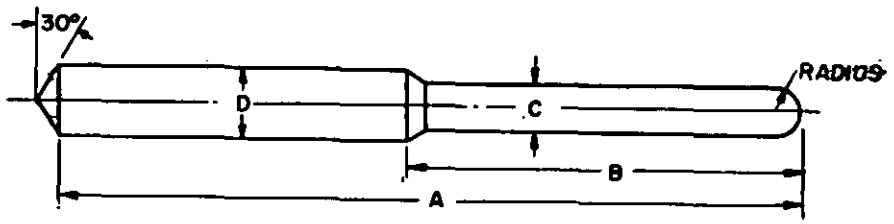


FIGURE 10. Expansion pins for type I, classes 2 and 3, puller kits, mechanical, bearing and bushing, removal, medium and heavy duty.

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TABLE III. Expansion pins for type I, classes 2 and 3 puller kit, mechanical, bearing and bushing, removal, medium and heavy duty

Size	Collet size nominal	Dimensions				
		A	B	C		D
		± 1/16	± 1/32	Minimum	Maximum	± 0.005
	Inches	Inches	Inches	Inch	Inch	Inch
1	1/4	2-11/16	1-1/2	0.093	0.100	3/16
	9/32	2-11/16	1-1/2	.093	.100	3/16
	5/16	2-11/16	1-7/16	.125	.135	3/16
	3/8	2-11/16	1-1/2	.147	.156	3/16
	7/16	2-11/16	1-1/2	.147	.156	3/16
	1/2	2-11/16	1-7/16	.170	.175	3/16
2	1/2 to 5/8	3-1/2	3/4	.224	.226	1/4
	5/8 to 3/4	3-5/8	3/4	.281	.285	5/16
	3/4 to 7/8	3-7/8	3/4	.281	.285	5/16
	7/8 to 1-1/16	4-1/8	Optional	.319	.313	5/16
	1-1/16 to 1-1/4	4-3/8-	Optional	.309	.313	5/16
	1-1/4 to 1-1/2	5-1/2	Optional	.465	.469	15/32
	1-1/2 to 1-3/4	5-3/4	Optional	.465	.469	15/32
	1-3/4 to 2	6	Optional	.465	.469	15/32

3.9.2.5 Adjusting screw. The adjusting screw shall be of good quality steel and shall be either casehardened to a depth of not less than 0.010-inch or heat treated to show not less than 40 on a Rockwell C scale. The adjusting screw shall be similar to figure 11,and shall conform to size 1 of table IV.

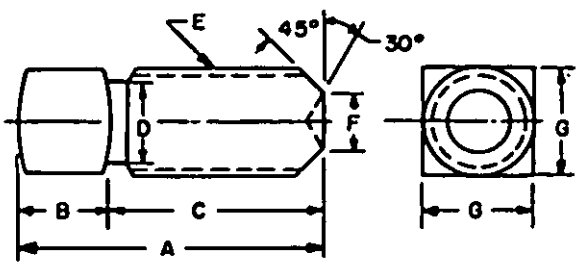


FIGURE 11. Adjusting screw for type I, classes 2 and 3, puller kit, mechanical, bearing and bushing, removal, medium and heavy duty.

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TABLE IV. Adjusting screw for type I, classes 2 and 3, puller kit, mechanical, bearing and bushing, removal, medium and heavy duty

Size	Dimensions						
	A	B	C	D	E	F	G
	± 1/16	± 1/32	Minimum	± 0.010	Thread UNC-2A	± 0.010	± 1/32
	Inches	Inch	Inches	Inch		Inch	Inch
1	1-3/16	5/16	7/8	0.300	3/8-16	1/4	3/8
2	2-1/8	5/8	1-1/2	.600	3/4-10	7/16	3/4

3.9.2.6 Collets. The collets shall be of high grade steel, heat treated to show not less than 40 nor more than 42 on a Rockwell C scale. The body of the collets shall have an external thread to accept the draw nut and the lower end of the collets shall be slotted to permit suitable expansion and retraction. Internally, the collets shall be suitably shaped to accept the expansion pins. The collets shall be similar to figures 12, 13, and 14, and shall conform to size 1 of table V.

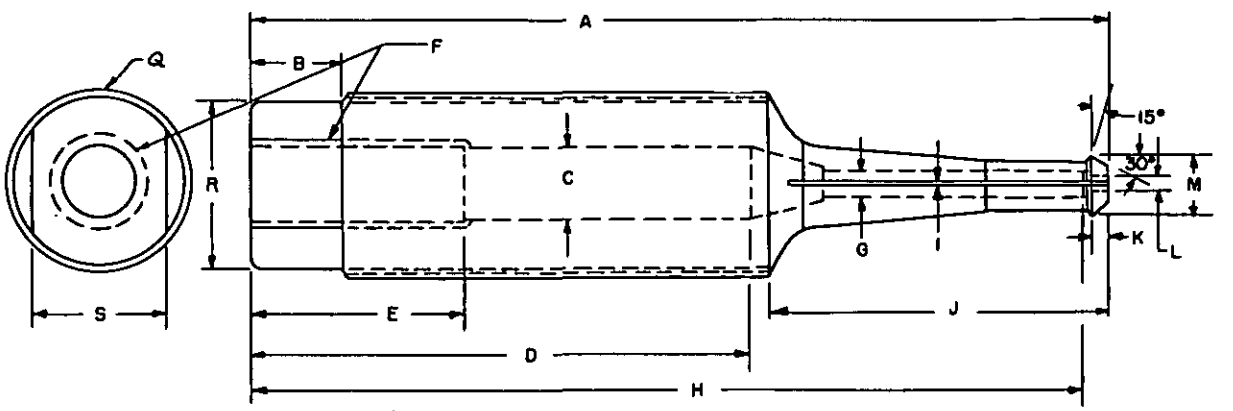


FIGURE 12. Collet for type I, classes 2 and 3 puller kit, mechanical bearing removal, medium and heavy duty.

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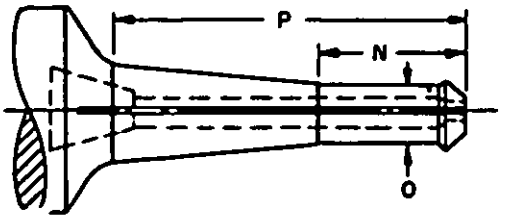


FIGURE 13. Expanding end of collet for type I, class 2 puller kit, mechanical, bearing and bushing removal, medium duty.

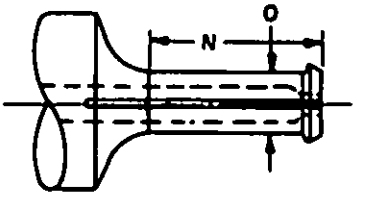


FIGURE 14. Expanding end of collet for type I, class 3 puller kit, mechanical bearing and bushing removal, heavy duty.

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TABLE V. Collets for type I, classes 4 and 3 puller kit, mechanical, bearing and bushing removal, medium and heavy duty

Size	Collet range			Collet dimensions																								
	Nom.	Min.	Max.	A	B	C	D	E	F	G	H	I		J	K	L	M		N	O		P	Q		R	S
				±1/16	±1/32	±0.005	±1/16	Min.	NC 2	±0.005	Max.	Width ±0.005	Length ±1/32	±1/32	±0.010	±0.005	Min.	Max.	±1/32	Min.	Max.	±1/32	Thread	±1/32	+0.000 -0.010	±0.005
	Inches	Inches	Inches	Inch	Inch	Inch	Inches	Inches	Thread	Inch	Inches	Inch	Inches	Inches	Inch	Inch	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inch
1	1/4	0.235	0.265	3-1/2	3/8	5/16	2-1/16	7/8	3/8-16	0.140	3-3/8	1/32	1-3/8	1-3/8	1/16	.078	0.248	0.252	1/2	0.205	0.210	1-1/4	3/4-16		11/16	9/16
	9/32	.265	.298	3-1/2	3/8	5/16	2-1/16	7/8	3/8-16	.140	3-3/8	3/64	1-3/8	1-3/8	1/16	.078	.279	.283	9/16	.245	.250	1-1/4	3/4-16		11/16	9/16
	5/16	.281	.343	3-1/2	3/8	5/16	2-1/16	7/8	3/8-16	.156	3-3/8	1/16	1-3/8	1-3/8	1/16	.093	.310	.314	1/2	.276	.281	1-1/4	3/4-16		11/16	9/16
	3/8	.343	.408	3-1/2	3/8	5/16	2-1/16	7/8	3/8-16	.177	3-3/8	3/64	1-3/8	1-3/8	0.100	.120	.373	.377	0.800	.330	.335	1-1/4	3/4-16		11/16	9/16
	7/16	.406	.468	3-1/2	3/8	5/16	2-1/16	7/8	3/8-16	.177	3-3/8	3/64	1-3/8	1-3/8	.100	.120	.435	.439	.800	.382	.387	1-1/4	3/4-16		11/16	9/16
	1/2	.470	.530	3-1/2	3/8	5/16	2-1/16	7/8	3/8-16	.200	3-3/8	1/16	1-3/8	1-3/8	.100	.140	.498	.502	.850	.445	.450	1-1/4	3/4-16		11/16	9/16
	9/16	1/2	5/8	4-7/8	1/2	21/32	2-5/8	1-1/2	3/4-10	9/32	4-3/4	1/16	1-5/8	1-3/4	3/32	.156	.560	.564	1-3/8	.486	.491	Not applicable		1-1/4-12	1-1/4	1
	11/16	5/8	3/4	5	1/2	21/32	3	1-1/2	3/4-10	11/32	4-59/64	1/16	1-3/4	1-3/4	7/64	13/64	.683	.687	1-3/8	.605	.610			1-1/4-12	1-1/4	1
	13/16	3/4	7/8	5-1/4	1/2	21/32	3-1/8	1-1/2	3/4-10	11/32	5-3/32	1/16	1-7/8	2	1/8	13/64	.810	.812	1-3/8	.727	.732			1-1/4-12	1-1/4	1
2	31/32	7/8	1-1/16	5-1/2	1/2	21/32	3-1/4	1-1/2	3/4-10	13/32	5-5/16	3/32	2	2	1/8	13/64	.964	.968	1-3/4	.855	.860			1-1/4-12	1-1/4	1
	1-5/32	1-1/16	1-1/4	5-3/4	1/2	21/32	3-3/8	1-1/2	3/4-10	13/32	5-1/2	3/32	2-1/8	2-1/8	1/8	13/64	1.152	1.156	1-3/4	.901	.906			1-1/4-12	1-1/4	1
	1-3/8	1-1/4	1-1/2	6-7/8	1/2	21/32	4	1-1/2	3/4-10	19/32	6-21/32	1/8	2-5/8	2-5/8	1/8	5/16	1.365	1.375	2-1/4	1.095	1.100			1-1/4-12	1-1/4	1
	1-5/8	1-1/2	1-3/4	7-1/8	1/2	21/32	4-1/4	1-1/2	3/4-10	19/32	6-29/32	1/8	2-5/8	2-5/8	1/8	5/16	1.615	1.625	2-1/4	1.095	1.100			1-1/4-12	1-1/4	1
	1-7/8	1-3/4	2	7-3/8	1/2	21/32	4-3/8	1-1/2	3/4-10	19/32	7-1/8	1/8	3-1/4	2-3/4	1/8	5/16	1.865	1.875	2-3/8	1.095	1.100			1-1/4-12	1-1/4	1

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3.9.2.7 Carrying case. A carrying case shall be provided to accommodate the puller kit parts with individual spaces. The case shall be of sheet steel not less than 0.024 inch thick and shall be of durable and rigid construction. The case shall be equipped with a hinged cover and one trunk latch. An instruction sheet containing clear and concise information as to the proper operation of the part shall be attached inside each carrying case.

3.9.3 Class 3, heavy duty. Class 3 puller kits shall be capable of removing bearings and bushings having inside diameters of not less than 1/2 nor more than 2 inches and shall consist of the same parts, as specified in 3.9.2. Class 3 puller kits shall be similar to figure 15.

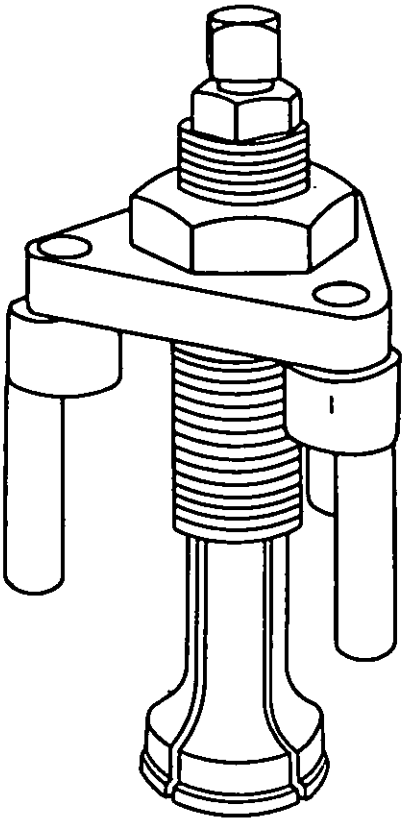


FIGURE 15. Type I, class 3 puller kit, mechanical, bearing and bushing, removal, heavy duty.

3.9.3.1 Draw nut. The draw nut shall be in accordance with the requirements of 3.9.2.1, except that the nut shall conform to the dimensions for size 2 of table I.

3.9.3.2 Thrust washer. The thrust washer shall be in accordance with the requirements of 3.9.2.2, except that the washer shall conform to the dimensions for size 2 of table II.

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3.9.3.3 Draw table. The draw table for the class 3 puller kit shall consist essentially of a table top and two sets of legs comprised of three short and three long legs. The table top shall be of good quality flat steel smoothly finished on both flat surfaces. Three holes at right angles to the flat surfaces shall be provided in the table top for accepting the legs. Means shall be provided in each hole to hold the legs in place under normal working conditions. The draw table shall conform to figure 16.

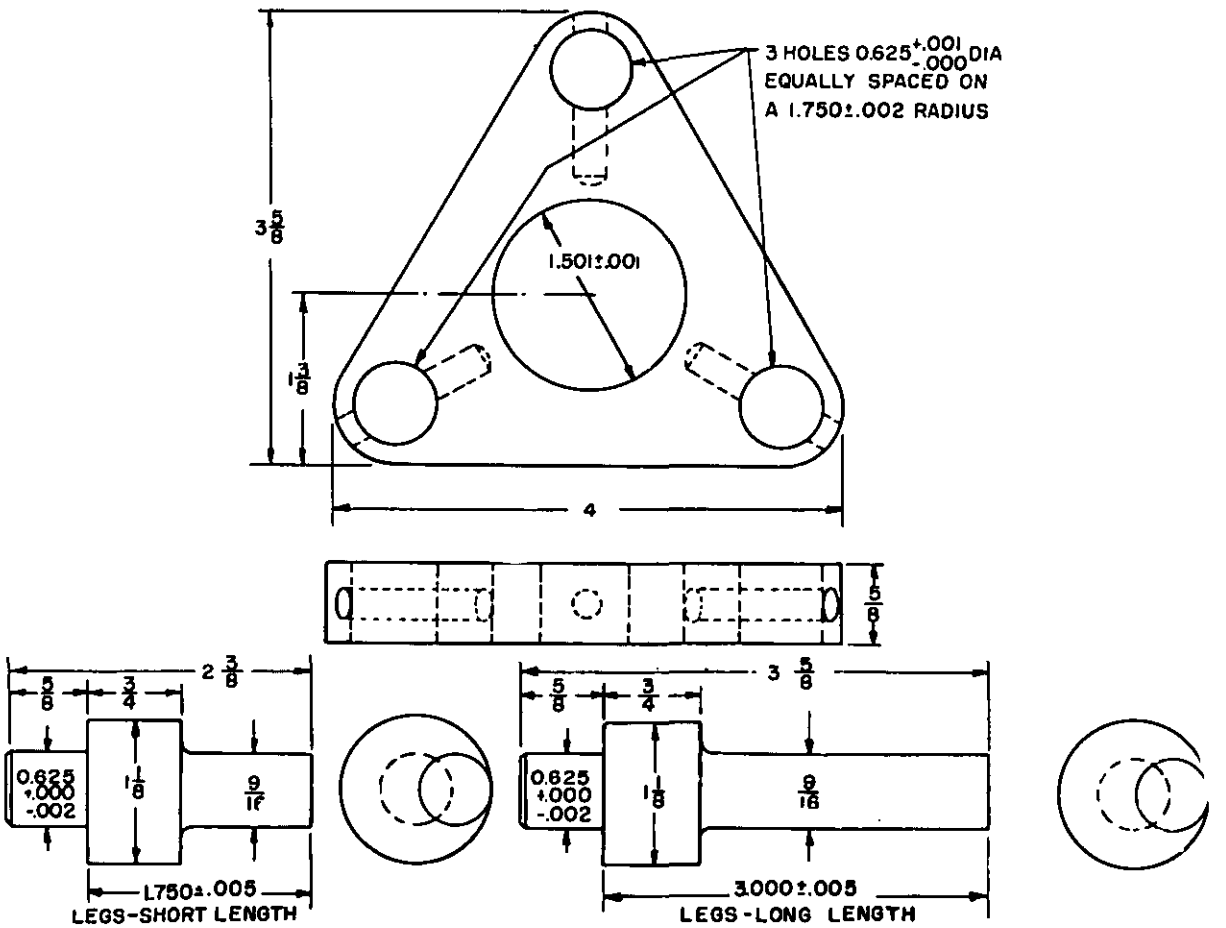


FIGURE 16. Draw table for type I, class 3 puller kit.

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3.9.3.4 Expansion pins. The expansion pins shall be in accordance with 3.9.2.4 except that the pins shall conform to size 2 of table III and shall be similar to figure 10.

3.9.3.5 Adjusting screw. The adjusting screw shall be in accordance with 3.9.2.5 except that the adjusting screw shall conform to size 2 of table IV and shall be similar to figure 11.

3.9.3.6 Collets. The collets shall be in accordance with 3.9.2.6 except that the collets shall conform to size 2 of table V. The collets shall be similar to figures 12 and 14 except that the collets are large in diameter.

3.9.3.7 Carrying case. A carrying case shall be provided to accommodate the puller kit parts with individual spaces. The case shall be of sheet steel not less than 0.030 inch thick and shall be of durable and rigid construction. The case shall be equipped with a hinged cover, two trunk latches, and a suitable carrying handle. An instruction sheet containing clear and concise information as to the proper operation of the part shall be attached inside each carrying case.

3.10 Type II, driver plug, bearing and bushing, installing and removing.

3.10.1 Class 1, engaging threaded adapter. The class 1 puller kit shall consist essentially of a shaft, housing, tension nut, tension nut handles, adapter sleeves, thrust bearing, and engaging threaded adapters. The parts shall be of good quality steel. The shaft and tension nut shall be properly heat treated to resist breakage or deformation.

3.10.1.1 Shaft. The shaft shall have one end threaded to engage the thread of the engaging threaded adapters and the body threaded to engage the thread of the tension nut. A groove shall be provided along the length of the body to insure clearance for the end of the pin.

3.10.1.2 Housing. The housing shall have a knurled portion and a smooth portion on the outer diameter. The internal diameters shall be suitable for accepting one end of the adapter sleeves and passage of the shaft. Means shall be provided at one end of the housing for insertion of a pin to prevent the shaft from turning.

3.10.1.3 Tension nut. The tension nut shall have a thread centrally located to engage the thread on the body of the shaft and threaded from both sides to engage the threads of the tension nut handles.

3.10.1.4 Tension nut handles. Handles shall be furnished for use with the puller kit. One end of all handles shall be provided with a thread to engage the threads in the sides of the tension nut.

3.10.1.5 Adapter sleeves. Adapter sleeves shall be designed in such a manner as to have the outer diameters suitable for entry into and butting against the end of the housing and an internal clearance hole suitable for passage of the bushing or bearing being removed.

3.10.1.6 Thrust bearing. A thrust bearing shall be provided for use with the puller kit and the bearing shall be of such size as to insure free movement of the nut during the entire installing and removing operations.

3.10.1.7 Engaging threaded adapters. Nine engaging threaded adapters shall be furnished for use with the puller kit. The adapters shall be so designed as to have an internal thread to engage the thread of the shaft and the outer diameter sized and flanged suitable for use in installing and removing bearings and bushings.

3.10.1.8 Carrying case. A carrying case shall be provided to accommodate the kit parts with individual holding spaces for the engaging threaded adapters. The case shall be sheet steel not less than 0.030 inch thick and shall be of durable and rigid construction. The case shall be equipped with a hinged cover, two trunk latches, and a suitable carrying handle. An instruction sheet containing clear and concise information as to the proper operation of the kit shall be attached inside each carrying case.

3.10.1.9 When puller attachment is required for use with class 1 puller kit, the puller attachment shall be in accordance with 3.13. The style puller kit for which it is intended shall be as specified (see 6.1).

3.10.2 Style A, light duty. Style A puller kit, when assembled with the proper adapter sleeve and engaging threaded adapter, shall be capable of installing and removing bearing and bushings having inside diameters of not less than 1/2 nor more than 1-1/16 inches. The kit shall be similar to figure 17 and shall withstand a minimum test load of 10,000 pounds.

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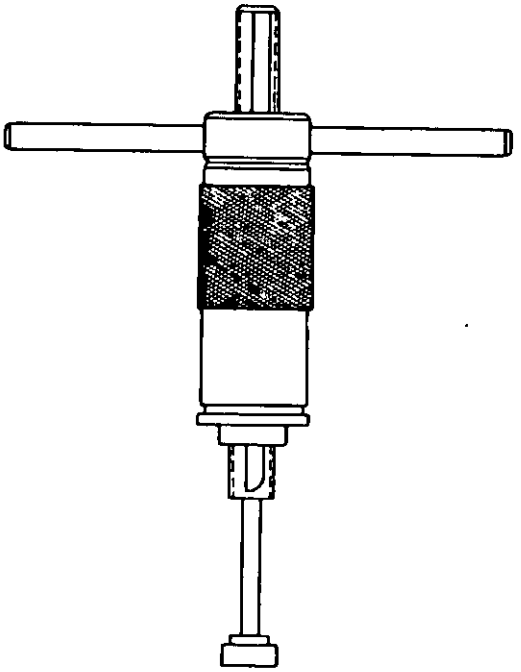


FIGURE 17. Type II, class 1, style A puller kit, mechanical, driver plug bearing and bushing, installing and removing, engaging threaded adapter, light duty.

3.10.2.1 Shaft. The shaft shall be in accordance with 3.10.1.1,and shall conform to the dimensions shown on figure 18.

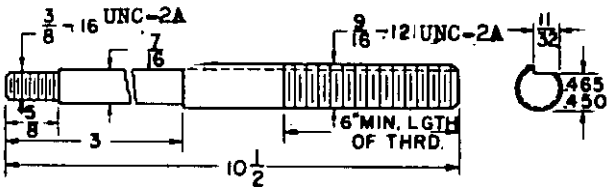


FIGURE 18. Shaft for type II, class 1, style A puller kits.

3.10.2.2 Housing. The housing shall be in accordance with 3.10.1.2, shall be similar to figure 19, and shall conform to size 1 of table VI.

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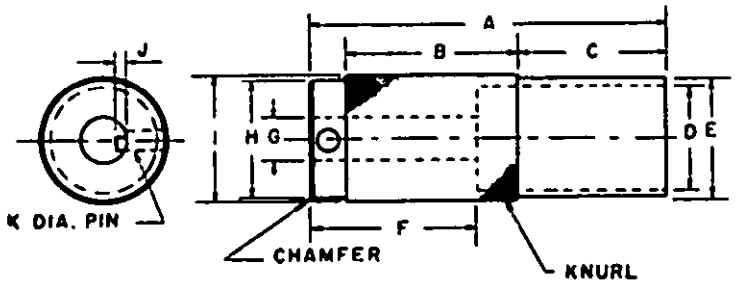


FIGURE 19. Housing for type II, class 1, styles A and B puller kits.

TABLE VI. Housing for type II, class 1, styles A and B puller kits

Size	Dimensions										
	A	B	C	D	E	F	G	H	I	J	K
	Inches	Inches	Inches	Inches	Inches	Inches	Inch	Inches	Inches	Inch	Inch
1	4-1/2	2-3/16	1-7/8	1-3/8	1-9/16	2-1/8	19/32	1-9/16	1-5/8	5/64	5/16
2	4-1/2	2-1/4	1-13/16	1-3/4	1-15/16	7/8	3/4	1-7/8	2	1/8	11/32

3.10.2.3 Tension nut. The tension nut shall be in accordance with 3.10.1.3, shall be similar to figure 20, and shall conform to size 1 of table VII.

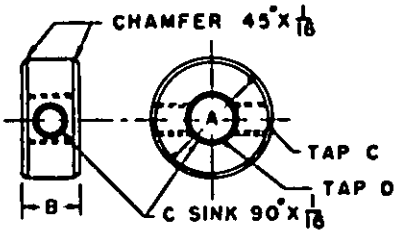


FIGURE 20. Tension nut for type II, class 1, styles A and B puller kits.

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TABLE VII. Tension nut for type II, class 1, styles A and B puller kits

Size	Dimensions			
	A	B	C	UNC-2B Thread D
	Inches	Inch	Inches	Inches
1	1-5/8	9/16	3/8-16 UNCA-2B	9/16-12
2	2	7/8	1/2-20 UNF-2B	3/4-10

3.10.2.4 Tension nut handle. The tension nut handle shall be in accordance with 3.10.1.4, shall be similar to figure 21, and shall conform to size 1 of table VIII.

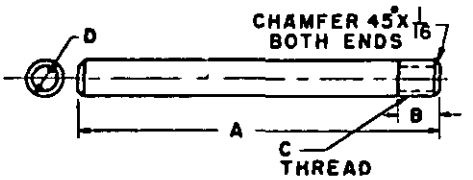


FIGURE 21. Tension nut handles, type II, class 1, styles A and puller kits.

TABLE VIII. Tension nut handles for type II, class 1, styles A and B puller kits

Size	Dimensions			
	A	B	C	D
	Inches	Inch	Inches	Inch
1	4-1/2	7/16	3/8-16 UNC-2A	3/8
2	5	9/16	1/2-20 UNF-2A	1/2

3.10.2.5 Adapter sleeves. Two adapter sleeves shall be furnished for use with style A puller kit, shall be in accordance with 3.10.1.5, shall be similar to figure 22, and shall conform to sizes 1 and 2 of table IX.

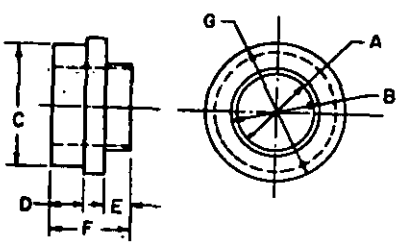


FIGURE 22. Adapter sleeves for type II, class 1, styles A and B puller kits.

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TABLE IX. Adapter sleeves for type II, class 1, styles and B puller kits

Size	Dimensions						
	A	B	C	D	E	F	G
	Inches	Inches	Inches	Inch	Inch	Inches	Inches
1	13/16	1.0	1-5/16	1/2	3/8	1-1/8	1-1/2
2	1-1/16	1-1/4	1-3/8	1/2	1/4	1.0	1-5/8
3	13/16	15/16	1-3/4	1/2	3/8	1-5/32	2
4	1-3/16	1-7/16	1-3/4	1/2	3/8	1-5/32	2
5	1-9/16	1-11/16	--	--	1/4	3	2

3.10.2.6 Engaging threaded adapter. Engaging threaded adapter shall be in accordance with 3.10.1.7, shall be similar to figure 23, and shall conform to table X.

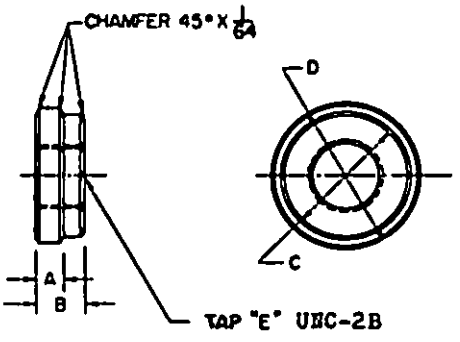


FIGURE 23. Engaging threaded adapter for type II, class 1, styles A and B puller kits.

TABLE X. Engaging threaded adapter for type II, class 1, style A puller kit

Engaging threaded adapter size	Dimensions				
	A	B	C +0.000 -0.010	D	E UNC-2B Thread
	Inch	Inch	Inches	Inches	Inches
1	5/32	1/4	1/2	5/8	3/8-16
2	5/32	1/4	9/16	3/4	3/8-16
3	5/32	1/4	5/8	13/16	3/8-16
4	5/32	1/4	11/16	7/8	3/8-16
5	5/32	1/4	47/64	15/16	3/8-16
6	5/32	1/4	51/64	1.00	3/8-16
7	5/32	1/4	15/16	1-1/8	3/8-16
8	5/32	1/4	1.0	1-3/16	3/8-16
9	5/32	1/4	1-1/16	1-1/4	3/8-16

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3.10.3.2 Housing. The housing shall be in accordance with 3.10.1.2, shall be similar to figure 19, and shall conform to size 2 of table VI.

3.10.3.3 Tension nut. The tension nut shall be in accordance with 3.10.1.3, shall be similar to figure 20, and shall conform to size 2 of table VII.

3.10.3.4 Tension nut handle. The tension nut handle shall be in accordance with 3.10.1.4, shall be similar to figure 21, and shall conform to size 2 of table VIII.

3.10.3.5 Adapter sleeves. Three adapter sleeves shall be furnished for use with style B puller kit, shall be in accordance with 3.10.1.5, shall be similar to figure 22, and shall conform to sizes 3, 4, and 5 of table IX.

3.10.3.6 Stud bolts. Four stud bolts shall be furnished for use with the style B puller kit. One end of the bolt shall be threaded to engage the small threaded hole of the reducing adapter and the opposite end threaded to engage the threaded hole in the engaging threaded adapter, shall be similar to figure 26, and shall conform to table XI.

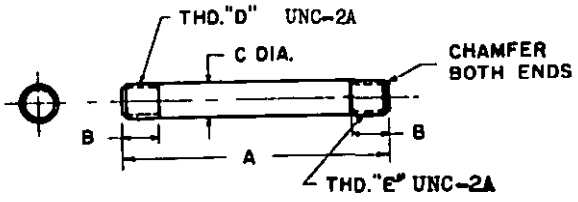


FIGURE 26. Stud bolts for type II, class 1, style B puller kit.

TABLE XI. Stud bolts for type II, class 1, style B puller kit.

Stud bolt size number	Dimensions				
	A	B	C	D UNC-2A Thread	E UNC-2A Thread
	Inches	Inches	Inch	Inches	Inches
1	3	7/16	1/4	1/4-20	1/4-20
2	3	1/2	3/8	3/8-16	3/8-16
3	3	1/2	3/8	1/4-20	3/8-16
4	3	1-1/2	3/8	3/8-16	3/8-16

3.10.3.7 Reducing adapter. Two reducing adapters shall be furnished for use with the style B puller kit. The adapter shall have in one end a small internally threaded hole for engaging the thread of one end of the stud bolts and the opposite end threaded to engage the thread of one end of the shaft. The reducing adapter shall conform to the dimensions of figure 27.

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3.10.2.7 Carrying case, light duty. The carrying case for the style A, light duty, kit shall be similar, in construction, to 3.10.1.8, except that latches and carrying handles are not required.

3.10.3 Style B, heavy duty. Style B puller kit, in addition to the parts specified in 3.10.1, shall consist essentially of stud bolts and reducing adapters, and shall be of good quality steel. Style B puller kit when assembled with the proper sleeve and engaging threaded adapter shall be capable of installing and removing bearings and bushings having inside diameters of not less than 29/64 inch nor more than 1-5/8 inch, shall be similar to figure 24, and shall withstand a minimum test load of 20,000 pounds.

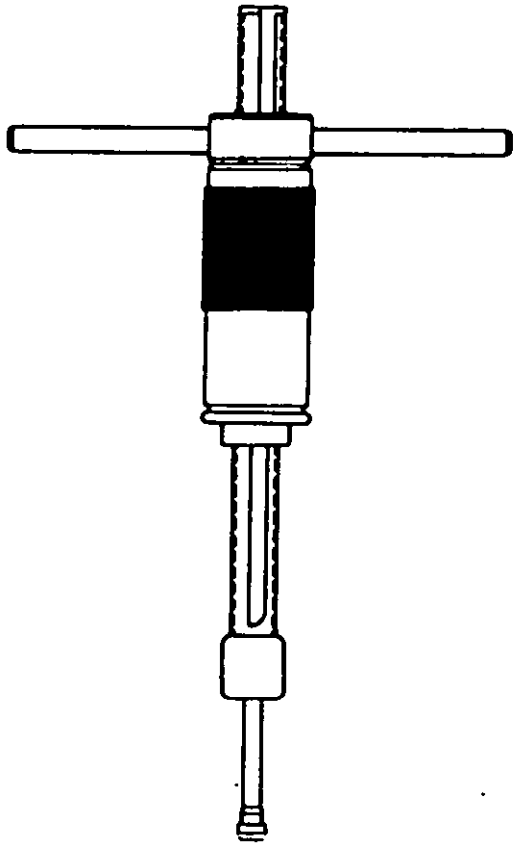


FIGURE 24. Type II, class 1, style B puller kit, mechanical, driver plug bearing and bushing, installing and removing, engaging threaded adapter, heavy duty.

3.10.3.1 Shaft. The shaft shall be in accordance with 3.10.1.1, and shall conform to the dimensions of figure 25.

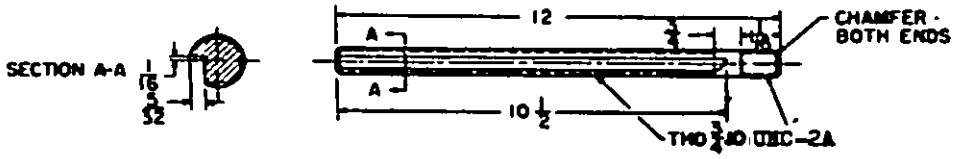


FIGURE 25. Shaft for type II, class 1, style B puller kit, mechanical.

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ADAPTER SIZE NO.	DIMENSION A
1	$\frac{1}{4}$ - 20
2	$\frac{3}{8}$ - 16

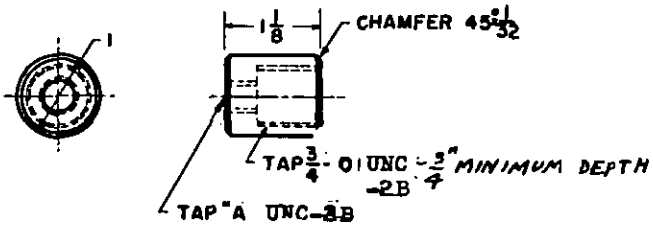


FIGURE 27. Reducing adapters for type II, class 1, style B, puller kit.

3.10.3.8 Engaging threaded adapters. Engaging threaded adapters shall be in accordance with 3.10.1.7, shall be similar to figure 23, and shall conform to table XII.

TABLE XII. Engaging threaded adapters for type II, class 1, style B puller kit

Engaging Threaded Adapter Size Number	Dimensions				
	A	B	C +0.000 -0.010	D	E UNC-2 Thread
	Inch	Inch	Inches	Inches	Inches
1	3/16	11/32	7/16	5/8	1/4-20
2	1/4	13/32	1/2	5/8	3/8-16
3	1/4	13/32	9/16	3/4	3/8-16
4	1/4	13/32	3/4		3/8-16
5	1/4	7/16	13/16	1.0	3/8-16
6	1/4	7/16	15/16	1-1/8	3/8-16
7	5/16	9/16	1-3/16	1-3/8	3/4-10
8	5/16	9/16	1-5/16	1-5/8	3/4-10
9	5/16	9/16	1-1/2	1-11/16	3/4-10

3.10.4 Class 2, engaging bushing with shaft. Class 2 puller kits shall consist essentially of a shaft, sleeves, sleeve ends, drivers, washer, nut, spring bushing shackle adapter, spring shackle puller, engaging bushings, and thrust bearing. All parts shall be of high grade steel and, except for the sleeves, the parts shall be properly heat treated to resist breakage or deformation. The puller kit when assembled with the proper sleeve and engaging bushing shall be capable of installing and removing bearings and bushings with inside diameters of not less than 13/32 nor more than 1-15/64 inches The class 2 puller kit shall be similar to figure 28, and shall withstand a minimum test load of 10,000 pounds.

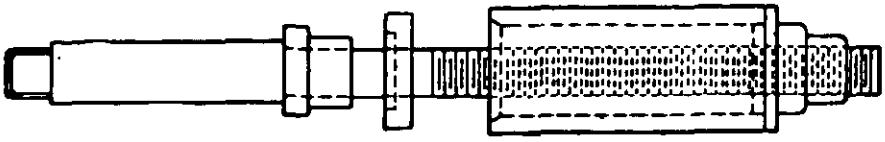


FIGURE 28. Type II, class 2 puller kit, mechanical, bearing and bushing, installing and removing, engaging bushing with shaft.

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3.10.4.1 When engaging bushings are required for use with type II, class 2 puller kit, the engaging bushings shall be in accordance with 3.10.4.10. The size or sizes of the engaging bushings shall be, as specified (see 6.1).

3.10.4.2 Shaft. The shaft shall be threaded to engage the thread in the nut and the opposite end having a square head drive of a size suitable for accepting a standard size wrench. The shaft shall conform to the dimensions shown on figure 29.

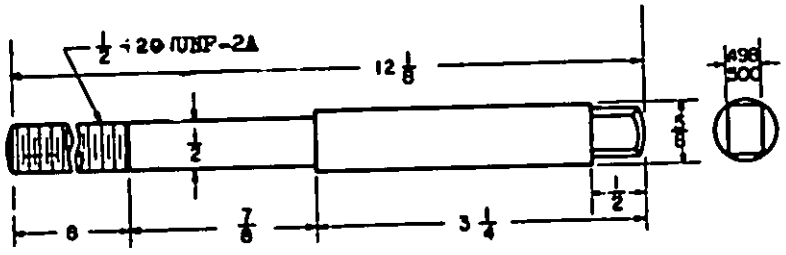


FIGURE 29. Shaft for type II, class 2 puller kit.

3.10.4.3 Sleeves. Three sleeves shall be furnished for use with type II, class 2 puller kit. The sleeves shall be similar to figure 30, and shall conform to table XIII.

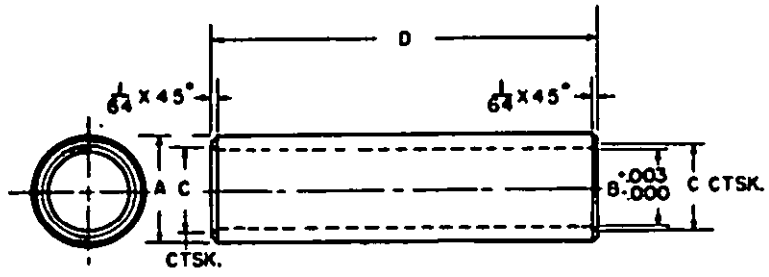


FIGURE 30. Sleeves for type II, class 2 puller kit.

TABLE XIII. Sleeves for type II, class 2 puller kit

Sleeve Size	Outside diameter	Inside diameter	Diameter of ctsk.	Length overall
	A	B	C	D
	Inches	Inches	Inches	Inches
1	1-5/16	1.015	1-1/16	4-15/16
2	1-11/16	1.376	1-27/64	4-15/16
3	1-11/16	1.376	1-21/64	3-3/16

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3.10.4.4 Sleeve ends. Two sleeve ends shall be furnished for use with type II, class 2 puller kit. Class 2 sleeve ends shall be similar to figure 31, and shall conform to table XIV.

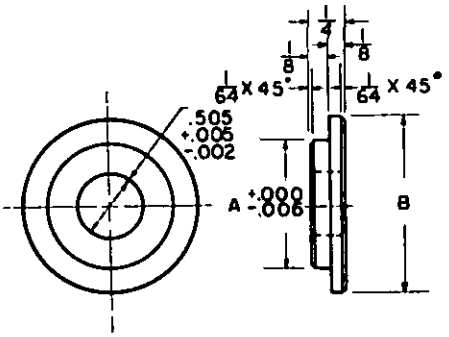


FIGURE 31. Sleeve ends for type II, class 2 puller kit.

TABLE XIV. Sleeve ends for type II, class 2 puller kit

Sleeve ends size number	Dimensions	
	A	B
	Inches	Inches
1	1.010	1-5/16
2	1.371	1-11/16

3.10.4.5 Drivers. Two drivers shall be provided for use with type II, class 2 puller kit. The driver shall be similar to figure 32, and shall conform to table XV.

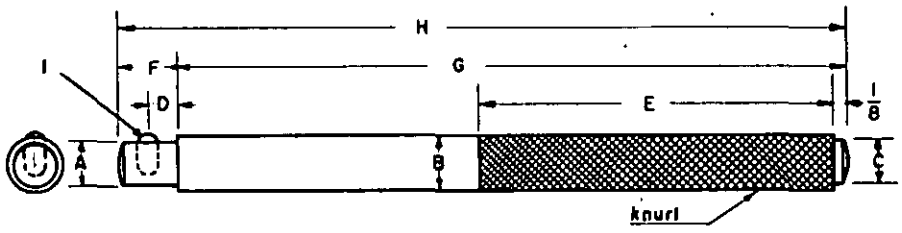


FIGURE 32. Drivers for type II, classes 2 and 3 puller kit.

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Table XV. - Drivers for type II, class 2 puller kit

Driver size number	Bushing locating end	diameter	Striking end	Location of ball and spring	Length of knurl	Length of locating end	Length of body	Overall length	Ball detent
	A	B	C	D	E	F	G	H	I
	Inch	Inch	Inch	Inch	Inches	Inch	Inches	Inches	Inch
1	0.310 to .312	7/16	3/8	1/4	2-5/8	1/2	4-1/4	4-3/4	1/8
2	0.498 to .500	5/8	9/16	5/16	3-7/8	5/8	7-3/8	8	3/16

3.10.4.6 Washer. The washer shall be suitable for use in installing bushings and bearings and shall conform to the dimensions shown on figure 33.

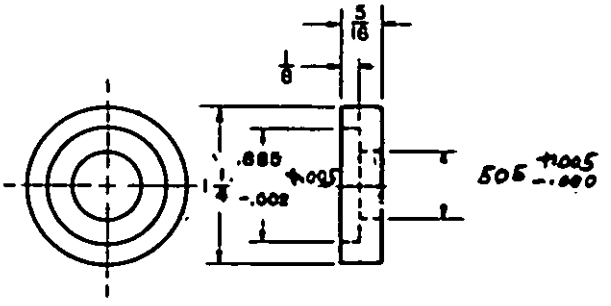


FIGURE 33. Washer for type II, classes 2 and 3 puller kit.

3.10.4.7 Nut. The nut shall be of appropriate size and threaded to engage the thread of the shaft. The thread shall conform to the dimensions shown on figure 34. The nuts shall be casehardened.

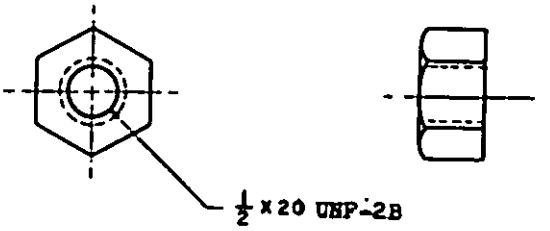


FIGURE 34. Nut for type II, class 2 puller kit.

3.10.4.8 Spring bushing shackle adapter. The adapter shall conform to the dimensions shown on figure 35.

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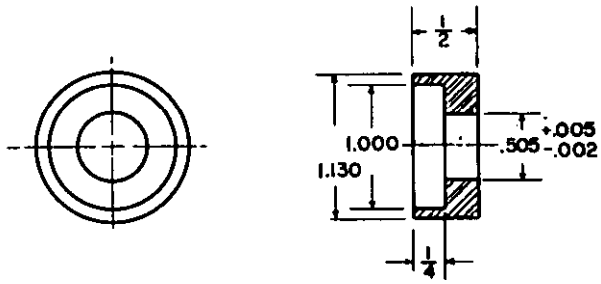


FIGURE 35. Spring bushing shackle adapter for type II, class 2 puller kit.

3.10.4.9 Spring shackle puller. The puller shall conform to the dimensions shown on figure 36.

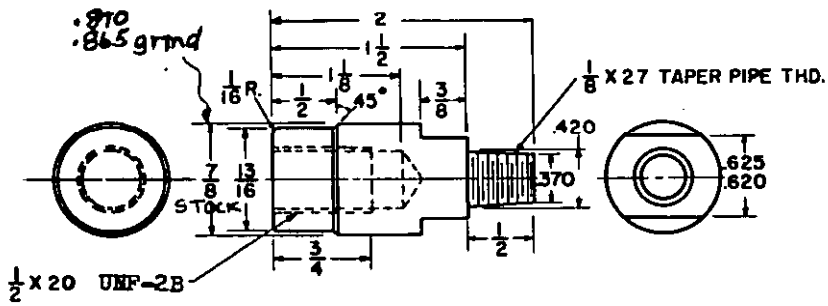


FIGURE 36. Spring shackle puller for type II, classes 2 and 3 puller kits.

3.10.4.10 Engaging bushing. Class 2 engaging bushings shall be similar to figure 37 and shall conform to table XVI.

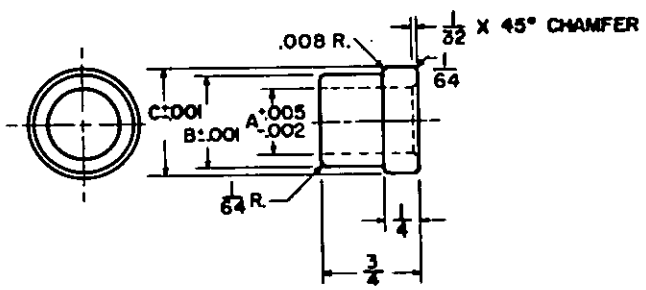


FIGURE 37. Engaging bushings for type II, class 2 puller kit.

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TABLE XVI. Engaging bushings for type II, class 2 puller kit

Size number	For bushing diameter	Inside diameter	Body diameter	Flange diameter
	I.D. O.D	A	B	C
	Inches	Inch	Inches	Inches
1	13/32 by 7/16	0.318	0.397	0.433
2	7/16 by 1/2	.318	.428	.496
3	1/2 by 9/16	.318	.491	.558
4	1/2 by 5/8	.318	.490	.620
5	9/16 by 11/16	.318	.552	.682
6	9/16 by 5/8	.505	.553	.621
7	39/64 by 43/64 5/8 by 11/16	.505 .505	.599 .599	.667 .667
8	5/8 by 3/4	.505	.616	.746
9	11/16 by 13/16 47/64 by 55/64	.505 .505	.678 .678	.808 .808
10	3/4 by 7/8	.505	.726	.855
11	3/4 by 1.0	.505	.740	.995
12	51/64 by 59/64 13/16 by 15/16 55/64 by 63/64	.505 .505 .505	.787 .787 .787	.917 .917 .917
13	7/8 by 1.0	.505	.849	.979
14	59/64 by 1-3/4 15/16 by 1-1/6	.505 .505	.905	1.045
15	0.927 by 1-1/32	.505	.917	1.026
16	1.0 by 1-1/8	.505	.991	1.120
17	1-1/16 by 1-3/16 1-7/64 by 1-15/64	.505 .505	1.053 1.053	1.183
18	1-1/8 by 1-1/4	.505	1.099	1.229
19	1.253 by 1.575 1-15/64 by 1-3/8	.505	1.220	1.370

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3.10.5.1 Bushing remover screws. Two bushing remover screws shall be furnished for use with type II, class 3 puller kit. The screw shall be similar to figure 38 and shall conform to table XVII.

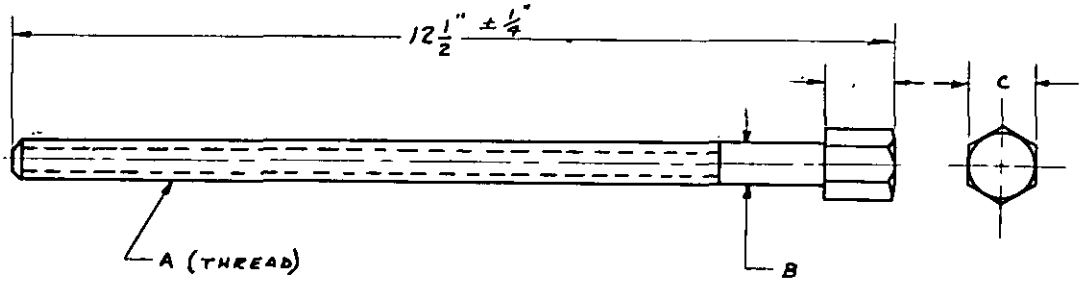


FIGURE 38. Bushing remover screws for type II, class 3 puller kit.

TABLE XVII. Bushing remover screws for type II, class 3 puller kit

A	"B" Diameter +.000 -.003	"C" Hexagon across flats
Thread	Inch	Inch
1/2-20 UNF-2A	0.500	5/8
5/8-18 UNF-2A	.625	1

3.10.5.2 Sleeve. One sleeve shall be furnished for use with type II, class 3 puller kit. The sleeve shall conform to the dimensions shown on figure 39.

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3.10.4.11 Thrust bearing. The thrust bearing shall be in accordance with 3.10.1.6.

3.10.4.12 Carrying case. The carrying case shall be in accordance with 3.10.1.8, except that suitable compartments shall be provided for the parts.

3.10.5 Class 3, engaging bushing with two screws. Class 3 puller kits shall consist of at least the following:

2 Bushing remover screws.
1 Sleeve.
3 Sleeve ends.
2 Drivers.
1 Washer.
2 Nuts.
1 Spring shackle puller.
2 Thrust bearings.
5 Bushing removers.
2 Adapters, idler-arm bushing removers.
1 Slotted washer.
1 Rear spring adaptor
22 Engaging bushings.
1 Carrying case.

All parts shall be of high steel; and except for the sleeve and slotted washer, the parts shall be properly heat treated to resist breakage or deformation. The class 3 puller kit shall be similar to figure 28 except that in lieu of the square-head drive shaft shown, two bushing removing screws with hexagon heads are substituted, and each screw with parts assembled shall withstand a minimum test load of 10,000 pounds.

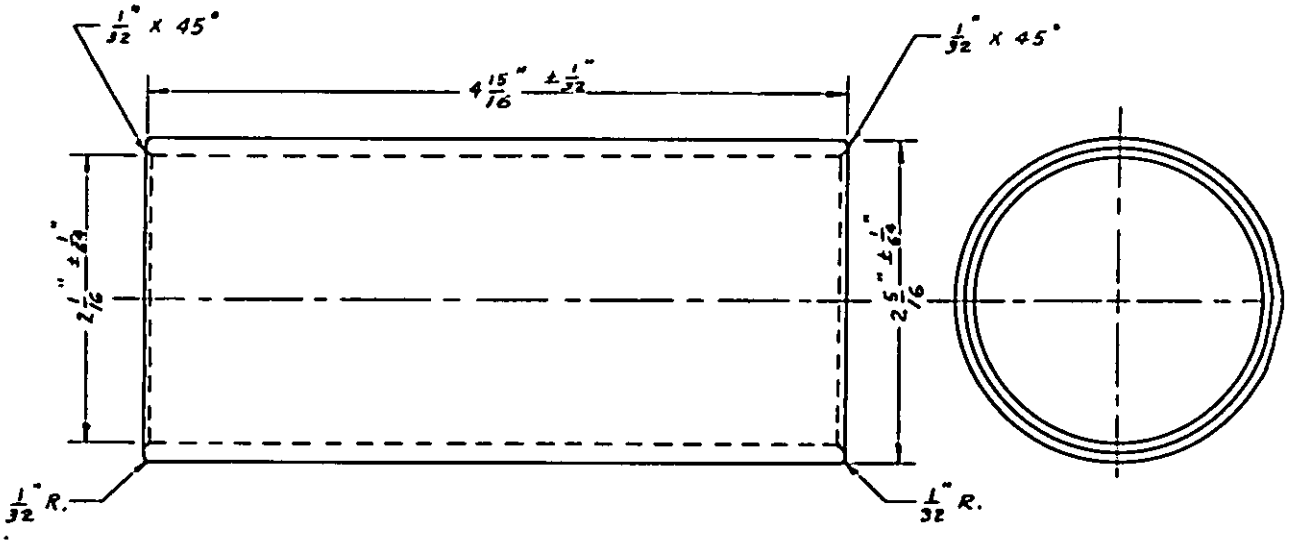


FIGURE 39. Sleeve for type II, class 3 puller kit.

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3.10.5.3 Sleeve ends. Three sleeve ends shall be furnished for use with type II, class 3 puller kits. The sleeve ends shall be similar to figure 40 and shall conform to the dimensions shown on figure 40 and in table XVIII.

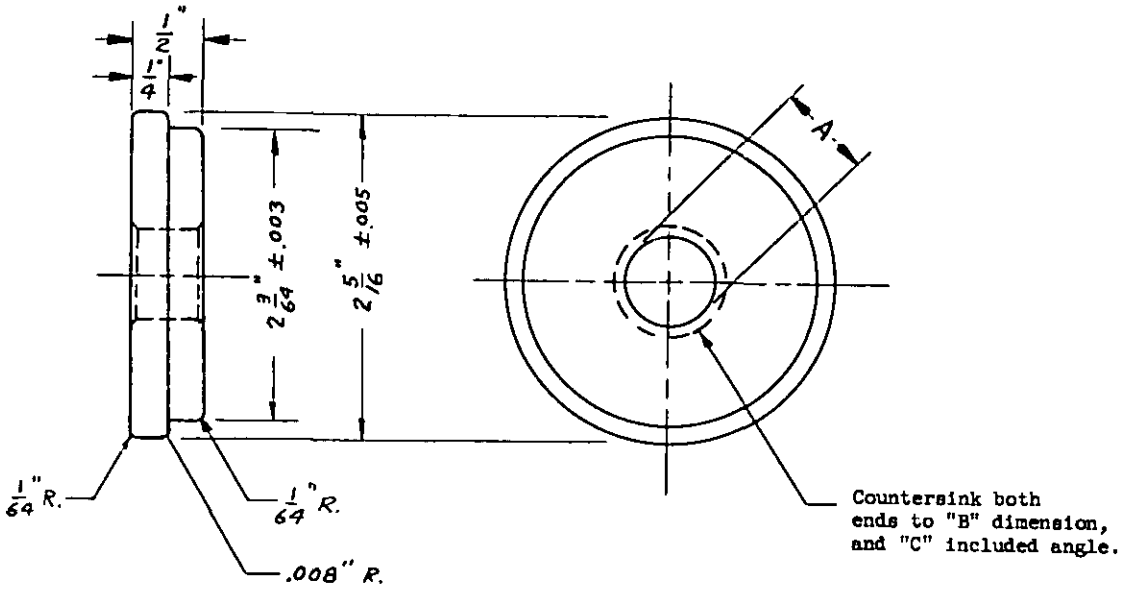


FIGURE 40. Sleeve end for type II, class 3 puller kit.

TABLE XVIII. Sleeve ends for type II, class 3 puller kit

Size No.	A - Diameter and tolerance	B	C
	Inches	Inches	Degrees
1	41/64 ± 0.005	45/64	120
2	1 ± .005	1-1/16	120
3	1-1/2 ± .005	1-17/32	90

3.10.5.4 Drivers. Two drivers shall be furnished for use with type II, class 3 puller kit. The drivers shall be similar to figure 32, and shall conform to the dimensions shown in table XIX.

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TABLE XIX. Drivers for type II, class 3 puller kit

Driver size number	Bushing locating end	Body diameter	Striking end	Location of ball and spring	Length of knurl	Length of locating end	Length of body	Overall length	Ball detent
	A	B	C	D	E	F	G	H	I
	Inch	Inch	Inch	Inch	Inches	Inch	Inches	Inches	Inch
1	0.307 to .312	7/16	3/8	1/4	2-5/8	1/2	4-1/4	4-3/4	1/8
2	0.495 to .500	5/8	9/16	5/16	3-7/8	5/8	7-3/8	8	3/16

3.10.5.5 Washer. The washer shall be suitable for use in installing bushings and bearings and shall conform to the dimensions shown on figure 33.

3.10.5.6 Nuts. Two nuts shall be furnished, one nut shall have a 5/8-18 UNF-2B thread and the other shall have a 1/2-20 UNF-2B thread. The nuts shall be casehardened.

3.10.5.7 Spring shackle puller. The puller shall conform to the dimensions shown on figure 36.

3.10.5.8 Thrust bearings. Two commercial thrust bearings shall be furnished. One thrust bearing shall have a 5/8-inch hole to fit over the large bushing remover screw, and the overall dimensions of the bearing shall be 11/32 inch thick by 1-1/8 inches in diameter. One thrust bearing shall have a 1/2-inch hole to fit over the small bushing remover screw, and the overall dimensions of the bearing shall be 11/32 inch thick by 1 inch in diameter.

3.10.5.9 Bushing removers. Five bushing removers shall be furnished. The bushing removers shall be similar to figure 41 and shall conform to table XX

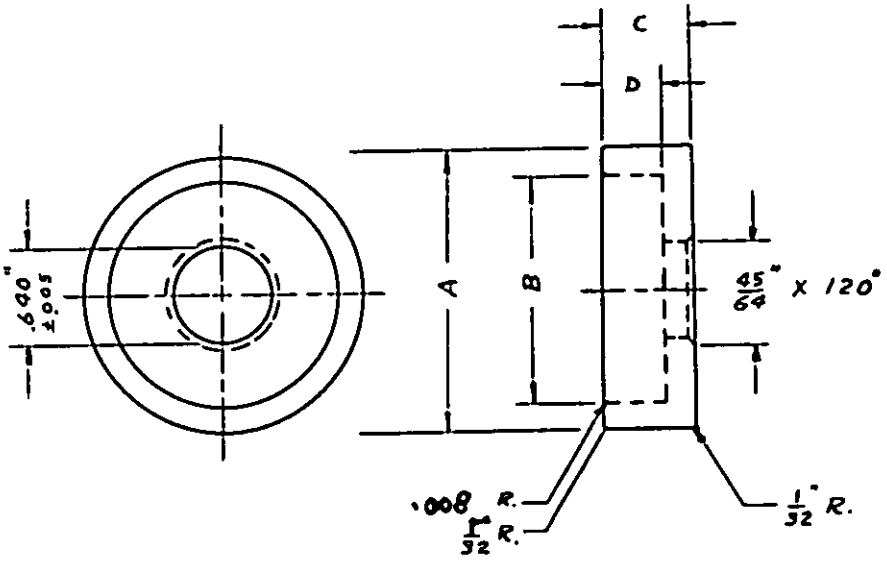


FIGURE 41. Bushing removers for type II, class 3 puller kit.

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TABLE XX. Bushing removers for type II, class 3 puller kits

Size No.	Outside diameter ±.005	Diameter ±.005	Thickness ±1/64	Depth ±1/64
	A	B	C	D
	Inches	Inches	Inch	Inch
1	1.250	1.125	7/16	9/32
2	1.406	1.187	1/2	5/16
3	1.812	1.375	9/16	3/8
4	1.990	1.860	1/2	1/4
5	2.000	1.625	5/8	7/16

3.10.5.10 Adapters. Two adapters for removing idler arm bushings shall be furnished. The adapters shall be similar to figure 42 and shall conform to the dimensions shown in table XXI.

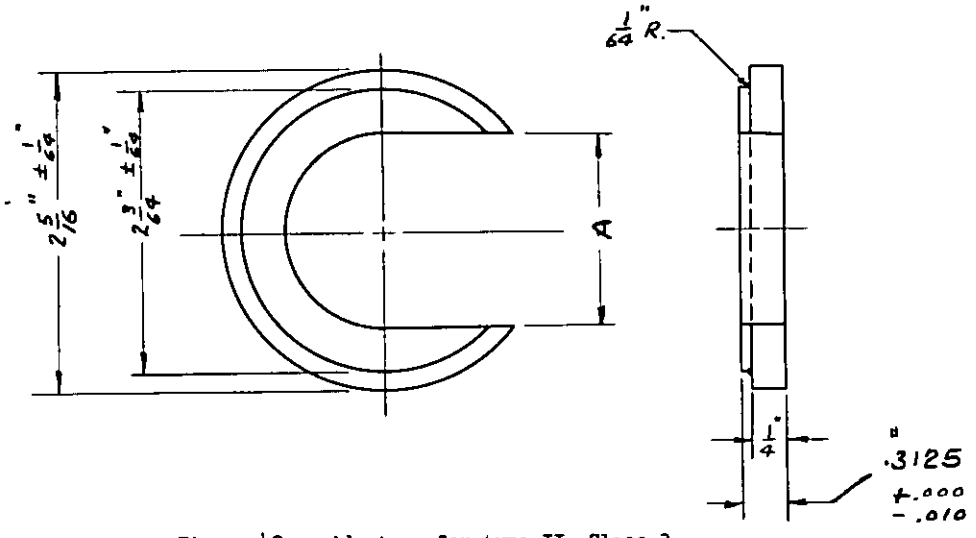


Figure 42. - Adapters for type II, Class 3 puller kit.

TABLE XXI. Adapters for type II, class 3 puller kit

Size Number	Slot size ± 1/64 A
	Inches
1	1-7/32
2	1-3/8

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3.10.5.11 Slotted washer. One slotted washer shall be furnished. The slotted washer shall be similar to figure 43 and shall conform to the dimensions shown.

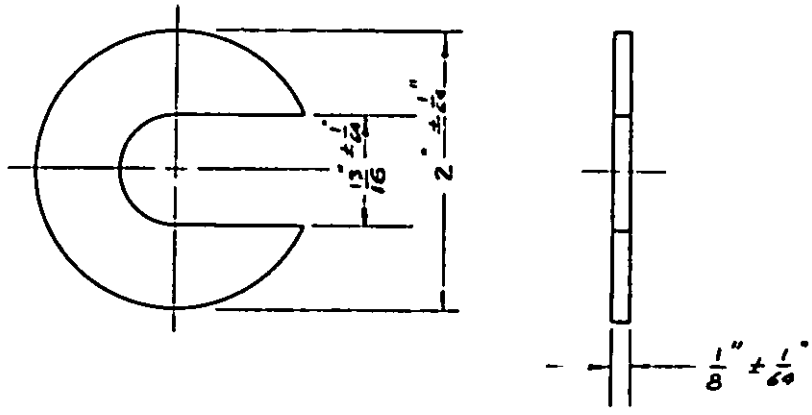


FIGURE 43. Slotted washer for type II, class 3 puller kit.

3.10.5.12 Rear spring adapter. One rear spring adapter shall be furnished. The adapter shall conform to the dimensions shown on figure 44.

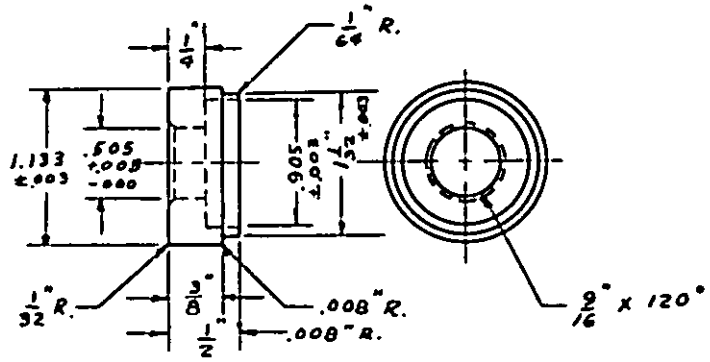


FIGURE 44. Rear spring bushing adapter.

3.10.5.13 Engaging bushings. Twenty-two engaging bushings shall be furnished. The bushings shall be in accordance with figure 45, except that the bushings shall be either radiused, chamfered, and/or countersunk as necessary to conform with good commercial practice. The bushings shall be in accordance with table XXII.

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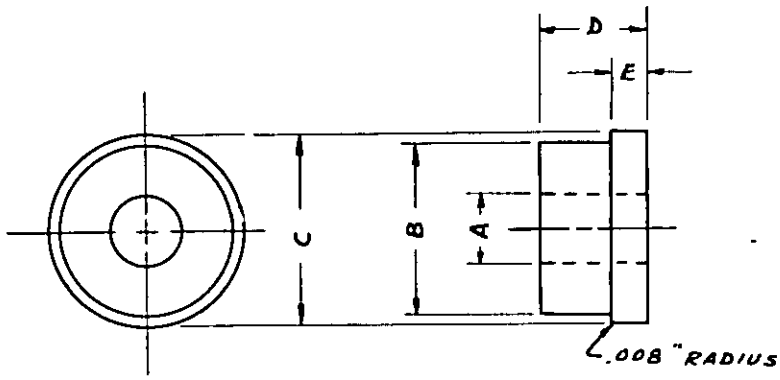


FIGURE 45. Engaging bushings, for type II, class 3 puller kits.

TABLE XXII. Engaging bushings for type II, class 3 puller kits

Size number	Inside diameter A	Body diameter B	Flange diameter C	Length D	Flange thickness E
	Inch	Inches	Inches		
1	0.318 +0.003 - .002	0.397 +0.002 - .003	0.432 +0.002 - .003	5/8	3/16
2	.318 +0.003 - .002	.428 +0.002 - .003	.495 +0.002 - .003	5/8	3/16
3	.318 +0.003 - .002	.491 +0.002 - .003 + .002	.557 +0.002 - .003 + .002	5/8	3/16
4	.318 0.002	.490 -0.003	.620 - .003	5/8	3/16
5	.318 +0.003 - .002	.552 +0.002 - .003	.682 +0.002 - .003	5/8	3/16
6	.505 +0.005 - .000	.553 +0.002 - .003	.620 +0.002 - .003	3/4	1/4
7	.505 +0.005 - .000	.599 +0.002 - .003	.666 +0.002 - .003	3/4	1/4
8	.505 +0.005 - .000	.616 +0.002 - .003	.745 +0.002 - .003	3/4	1/4
9	.505 +0.005 - .000	.678 +0.002 - .003	.807 +0.002 - .003	3/4	1/4
10	.505 +0.005 - .000	.726 +0.002 - .003	.854 +0.002 - .003	3/4	1/4
11	.505 +0.005 - .000	.740 +0.002 - .003	.995 +0.002 - .003	3/4	1/4
12	.505 +0.005 - .000	.787 +0.002 - .003	.916 +0.002 - .003	3/4	1/4
13	.505 +0.005 - .000	.849 +0.002 - .003	.978 +0.002 - .003	3/4	1/4
14	.505 +0.005 - .000	.905 +0.002 - .003	1.044 +0.002 - .003	3/4	1/4
15	.505 +0.005 - .000	.917 +0.002 - .003	1.025 +0.002 - .003	3/4	1/4
16	.505 +0.005 - .000	.991 +0.002 - .003	1.120 +0.002 - .003	3/4	1/4
17	.505 +0.005 - .000	1.053 +0.002 - .003	1.182 +0.002 - .003	7/8	1/4
18	.505 +0.005 - .000	1.099 +0.002 - .003	1.228 +0.002 - .003	3/4	1/4
19	.505 +0.005 - .000	1.220 +0.002 - .003	1.370 +0.002 - .003	3/4	1/4
20	.505 +0.005 - .000	.625 +-0.016	1.125 +- .016	3/4	1/4
21	.641 +-0.005	1.203 +-0.003	1.500 +- .016	7/8	3/8
22	.641 +-0.005	1.364 +-0.003	1.625 +- .016	7/8	3/8

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3.10.5.14 Carrying case. A carrying case shall be provided to accommodate the kit parts. The case shall be of sheet steel. The case shall be provided with a cover and latch.

3.11 Type III, cylinder sleeve, installing and removing. Type III puller kits shall consist essentially of a three-way head, forcing screw, forcing nut, riser block, thrust washer, thrust bearing, swivel assembly, forcing-screw extension, yoke assembly, crossbar, pulling adapter plates, and installing plates. All parts except the pulling adapter and installing plates, shall be properly heat treated to resist breakage or deformation. The type III puller kit without adapter plates shall be similar to figure 46.

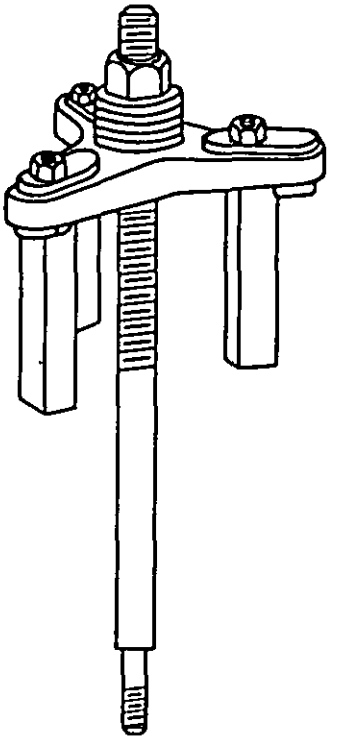


FIGURE 46. Type III puller kit, mechanical, cylinder sleeve, installing and removing.

3.11.1 Capacity. The puller kit, when assembled with the proper set of riser blocks and pulling adapter plates or installing plates, shall be capable of pulling cylinder sleeves (wet or dry) with inside diameters of not less than 3 inches nor more than 7-1/4 inches, and shall withstand a minimum test load of 10,000 pounds.

3.11.2 Pulling adapter plates and installing plates for use with type III puller kit shall be in accordance with 3.11.13 and 3.11.14, respectively. The size or sizes of the pulling adapter plates (wet or dry or both) and installing plates shall be, as specified (see 6.1).

3.11.3 Head. The head shall be of high grade plate or cast ductile steel and designed to have three slots equally spaced around the periphery of the head. The slots shall be of a size suitable for passage of the bolts in the securing of the riser blocks at the desired range. A smooth clearance hole, centrally located shall be provided in the head for passage of the forcing screw.

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3.11.4 Forcing screw. The forcing screw shall be of high grade alloy steel. The forcing screw shall be approximately 31 inches in length, having a 1-14 UNS-2A or 1-8UNC-2A thread, approximately 21 inches long from one end and suitable means for accepting and the attaching of the swivel shall be provided on the opposite end.

3.11.5 Forcing nut. The nut shall be of good quality steel, hexagonal in shape, and threaded to engage the thread of the forcing screw.

3.11.6 Riser blocks. The blocks shall be of high grade alloy steel, rectangular or cylindrical in cross section. One set, three blocks to a set, 13 inches in length shall be furnished for use with the puller kit. One end of the blocks shall have an internal thread to engage the thread of the screws. Attaching cap screws and washers shall be supplied.

3.11.7 Thrust washer. The thrust washer shall be of good quality steel and shall be suitable for use between the thrust bearing and the head.

3.11.8 Thrust bearing. The thrust bearing shall be in accordance with 3.10.1.6.

3.11.9 Swivel assembly. The swivel assembly shall consist essentially of an adapter and swivel which shall be of high grade alloy steel. The adapter shall be designed to engage the threaded end of the forcing screw and to engage the swivel. The swivel shall be designed to engage one end of the adapter and shall be properly secured to the adapter by means of a capscrew and nut. The design of the swivel shall also provide for the engaging and the securing of the pulling adapter plates. The swivel assembly shall be similar to figure 47.



FIGURE 47. Swivel assembly for type III puller kit.

3.11.10 Forcing-screw extension. The forcing-screw extension shall be of high grade alloy steel designed for use in connection with the forcing screw to engage the yoke assembly. One end of the extension shall engage the small threaded end of the forcing screw and on the opposite end, a thread shall be provided suitable for engaging the yoke assembly. The forcing-screw extension shall be similar to figure 48.



FIGURE 48. Forcing-screw extension for type III puller kit.

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3.11.11 Installing yoke assembly. The installing yoke assembly shall consist of an adapter and saddle and shall be of good quality steel. The adapter shall be designed to engage the thread on the forcing-screw extension and to engage the opening in the saddle. The saddle shall be designed to be secured to the adapter by a pin or other suitable means and shall have an enclosure suitable for nesting the crossbar. The yoke assembly shall be similar to figure 49.



FIGURE 49. Installing yoke assembly for type III puller kit.

3.11.12 Installing crossbar. The installing crossbar shall be of good quality steel. The crossbar shall be approximately 3/4 inch in thickness by 4 inches in width and 20 inches in length and shall be similar to figure 50.



FIGURE 50. Installing crossbar for type III puller kit.

3.11.13 Pulling adapter plates. The adapter plates shall be of cast iron, ductile steel or steel plate. The adapter plates shall be similar to figure 51 and shall conform to table XXIII or XXIV.

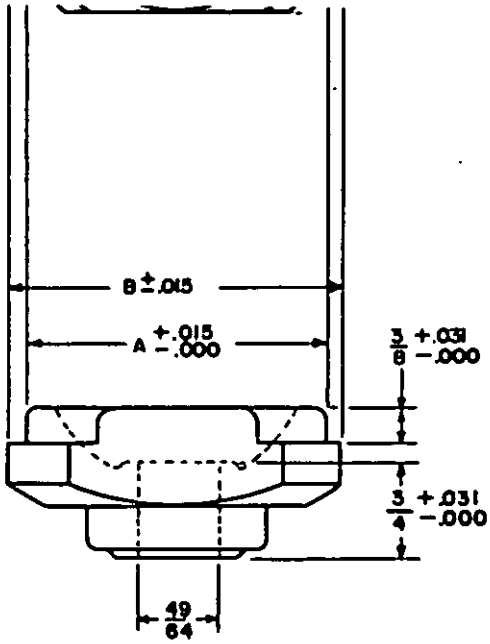


FIGURE 51. Pulling adapter plates for use with type III puller kit.

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TABLE XXIII. Pulling adapter plates for use with type III puller kit, wet cylinder sleeve, removal

Size number	For cylinder sleeve bore	Adapter plate	
		A	B
	Inches	Inches	Inches
1	3.0	2-31/32	3-11/32
2	3.0	2-63/64	3-11/64
3	3-1/8	3-3/32	3-15/32
4	3-1/16	3-5/32	3-17/32
5	3-1/4	3-7/32	3-19/32
6	3-5/16	3-19/64	3-43/64
7	3.314	3-19/64	3-5/8
8	3-3/8	3-11/32	3-23/32
9	3-7/16	3-27/64	3-51/64
10	3-1/2	3-15/32	3-13/16
11	3-5/8	3-19/32	3-31/32
12	3-7/8	3-27/32	4-3/16
13	4.0	3-31/32	4-13/32
14	4-1/8	4-3/32	4-17/32
15	4-1/4	4-7/32	4-21/32
16	4-3/8	4-11/32	4-25/32
17	4.440	4.390	4-39/64
18	4-1/2	4-15/32	4-29/32
19	4-5/8	4-19/32	5-1/64
20	4-3/4	4-23/32	5-7/32
21	4-7/8	4-27/32	5-11/32
22	5.0	4-31/32	5-15/32
23	5-1/8	5-3/32	5-19/32
24	5-1/4	5-7/32	5-23/32
25	5-3/8	5-11/32	5-27/32
26	5-1/2	5-15/32	5-31/32
27	5-5/8	5-19/32	6-3/32
28	5-3/4	5-23/32	6-1/4
29	5-7/8	5-27/32	6-11/32
30	6.0	5-31/32	6-15/32
31	6-1/4	6.235	6-39/64
32	6-3/4	6.734	7-1/4
33	7	6-63/64	7-21/32
34	7-1/4	7-15/64	7-3/4

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TABLE XXIV. Pulling adapter plates for use with type III puller kit, dry cylinder sleeve removal

Size number	For cylinder sleeve bore	Adapter plates	
		A	B
	Inches	Inches	Inches
1	3-3/16	3.171	3-11/32
2	3-1/4	3.238	3.360
3	3-3/8	3-23/64	3-35/64
4	3-9/16	3-35/64	3-51/64
5	3-5/8	3.610	3.778
6	3-3/4	3-47/64	4-3/32
7	3-3/4	3-47/64	3-63/64
8	3-7/8	3-55/64	4-3/32
9	4.0	3-63/64	4-15/64
10	4-1/8	4-7/64	4-31/64
11	4-3/16	4-11/64	4-7/16
12	4-1/4	4.235	4.470
13	4-3/8	4-23/64	4-31/64
14	4-1/2	4.485	4.937
15	4-5/8	4-39/64	4-55/64
16	4-3/4	4-47/64	5-11/64
17	4-13/16	4.797	4.906
18	4-7/8	4.860	4.980

3.11.14 Installing adapter plates. The installing adapter plates shall be of either steel plate or ductile steel. The installing plates shall be similar to figure 52 and shall conform to table XXV.

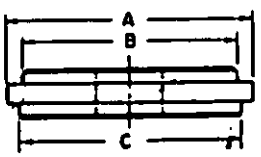


FIGURE 52. Installing adapter plates for type III puller kit.

TABLE XXV. Installing adapter plates for use with type III puller kit

Size number	For cylinder sleeves		Outside diameter	Diameter	Diameter
	From	To	A	B	C
	Inches	Inches	Inches	Inches	Inches
1	3-3/16	3-7/16	3-7/8	3.171	3.421
2	3-9/16	3-7/8	4-1/4	3.546	3.859
3	4-1/8	4-3/8	4-3/4	4.109	4.359
4	4-3/8	4-3/4	5-1/4	4.390	4.734
5	4-13/16	5-3/4	6-3/8	4.797	5.734

3.11.15 Carrying case. A carrying case, in accordance with 3.10.5.14, shall be provided.

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3.12 Type IV wheel hub. Type IV puller kit shall be designed for use in pulling wheel hubs, having grooved hubs or studs,and capable of pulling hubs up to a 9-inch bolt circle. The set shall consist essentially of a hub, forcing screw, drive handle, jaws, suitable grooved hub-pulling arrangement, nuts, and carrying case. All parts shall be properly heat treated to resist breakage or deformation.

3.12.1 Hub. The hub shall be of high grade alloy steel. The hub shall be designed to provide proper assembly and adjustment of the jaws around the periphery of the hub and the center hole shall be threaded or a centrally located nut provided to engage the thread of the forcing screw.

3.12.2 Forcing screw. The forcing screw shall be of high grade alloy steel and threaded to engage the thread in the hub or hub nut without binding or excessive play. Means shall be provided at the head of the screw for accepting the drive handle and the opposite end shall be suitably shaped for protection of the thread. The length of the screw, excluding the head, shall be equal to or slightly exceed the reach dimension plus the thickness of the hub.

3.12.3 Drive handle. The drive handle shall be of high grade forged alloy steel. The handle shall be permanently attached to the head of the forcing screw and shall be appropriately shaped to withstand the blows of a heavy striking object. The outer ends of the handle shall be suitably shaped for the hammering operation.

3.12.4 Jaws. Type IV hub pullers shall be furnished with four short jaws for pulling hubs having a 7-inch bolt circle, or three long jaws for pulling hubs having a 9-inch bolt circle, or both, as specified (see 6.1). The four jaws for the 7-inch bolt circle shall have 5/8 or 11/16 inch diameter bolt holes countersunk to accept conical nuts. The three jaws for the 9-inch-bolt circle shall have 13/16-inch diameter bolt holes countersunk to accept conical nuts. The jaws shall be of high grade forged alloy steel. The upper portion of the jaws shall be designed in a manner to prevent the jaws from slipping off the hub and to be easily assembled and removed. The lower portion of the jaws shall be properly proportioned with the gripping ends suitable for accepting the wheel hub studs in a manner as to facilitate a straight pull.

3.12.5 The puller shall be capable of pulling a 56,000-pound test load and shall be similar to figure 53.

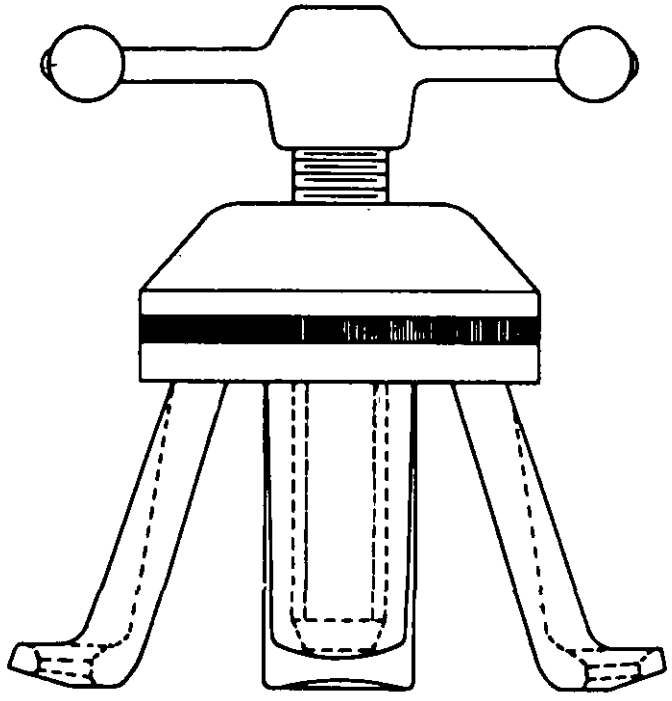


FIGURE 53. Hub-pulling arrangement for type IV puller kit.

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3.12.6 Stud nuts. The stud nuts shall be of high grade steel, hexagonal in shape, and shall be threaded to engage the threads of the wheel-hub studs.

3.12.6.1 Short stud nuts. As specified (see 6.1), two sets, four nuts to a set of size 2, or one set each, four nuts to a set of sizes 1 and 3, shall be furnished with the puller kit. Nuts shall be similar to figure 54 with conical taper, and sizes shall conform to table XXVI.

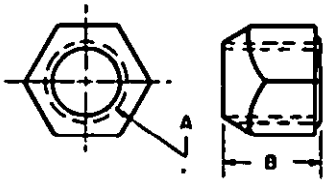


FIGURE 54. Short stud nuts for type IV puller kit.

TABLE XXVI. Short stud nuts for type IV puller kit

Size	Thread A UNF-2B	Nut length B
	Inch	Inch
1	7/16 - 20	13/16
2	1/2 - 20	13/16
3	9/16 - 18	13/16

3.12.6.2 Long stud nuts. One set, three nuts to a set, shall be furnished with the puller kit and shall conform to the dimensions of figure 55.

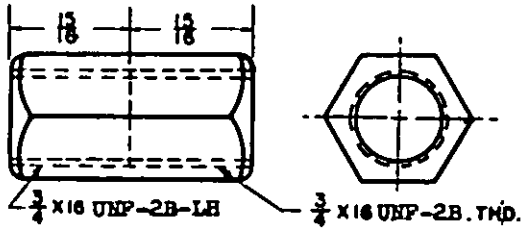


FIGURE 55. Long stud nuts for type IV puller kit.

3.12.6.3 Shaft protector. Shaft protector shall be provided meeting the requirements of GGG-P-781, Type III, class 3, size 3.

3.12.6.4 Carrying case. The carrying case shall be constructed of sheet steel not less 18 gage (0.0478) thickness. The cover shall be hinged by either a piano or form-type hinge. The case shall be divided into compartments to adequately hold the parts of the puller kit. The case shall be of welded construction and shall be provided with a metal carrying handle. All surfaces of the case shall be free from sharp edges, burrs, scale and painting defects. The case shall be painted, as specified in 3.14.2.

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3.13 Puller attachment, bearing and bushing installing and removing kit, mechanical, type V. Type V puller attachment when assembled with type II, class 1, styles A and B puller kits shall be capable of removing blind hole bearings and bushings having inside diameters of not less than 1/2 inch nor more than 1-1/2 inches, and shall consist essentially of an internal puller, sleeve, and sleeve adapter.

3.13.1 Internal puller. The puller shall consist of a yoke, reducing adapter and jaws, and shall conform to the dimensions shown on figure 56.

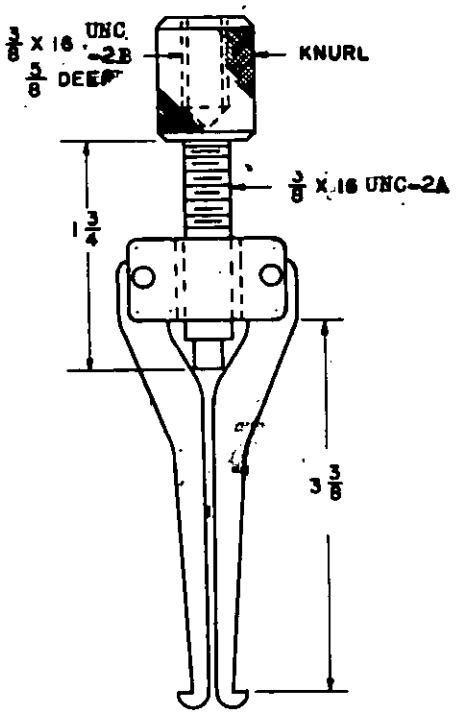


FIGURE 56. Internal puller for puller attachment, for use with type II, class 1, styles A and B puller kits.

3.13.1.1 Yoke. The yoke shall be of good quality steel and have a threaded hole centrally located to engage the thread of the reducing adapter. The jaws shall be asembled directly to the yoke and in parallel alignment.

3.13.1.2 Reducing adapter. The reducing adapter shall be of good quality steel. The head of the adapter shall have an internal thread to engage the thread of the shaft and the body threaded to engage the thread of the yoke. The opposite end of the adapter shall be turned to the root of thread and shall be suitably shaped for expanding the jaws.

3.13.1.3 Jaws. The jaws shall be of high grade forged steel. The gripping ends shall be properly proportioned to assure a firm grip on all pulling operations.

3.13.2 Sleeve. The sleeve shall be of good quality steel tubing. The sleeve shall conform to the dimensions shown on figure 57.

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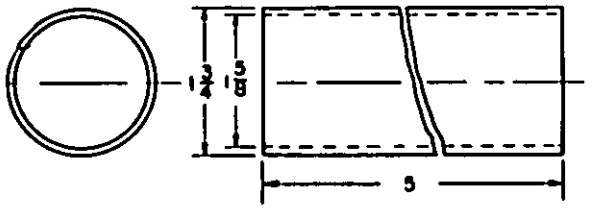


FIGURE 57. Sleeve, for puller attachment for use with type II, class 1, styles A and B puller kits.

3.13.3 Sleeve adapter. The sleeve adapter shall be of good quality steel. The adapter shall conform to the dimensions shown on figure 58.

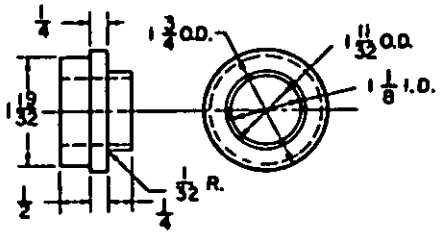


FIGURE 58. Sleeve adapter for puller attachment for use with type II, class 1, styles A and B puller kits.

3.14 Finish.

3.14.1 Surface. All puller kit components shall be substantially free from pits, nodules, burrs, cracks, and other detrimental defects. Surfaces not usually ground or buffed shall have a surface finish conforming to the best manufacturing practices.

3.14.2 Coating. The coating shall be adherent, smooth, continuous, and free from uncoated areas, pits, blisters, nodules, and any other defects which would interfere with the protective value and serviceability. Each case, furnished with a kit, shall be coated inside and outside with a suitable primer and not less than one coat of high-quality enamel. Forcing screws shall be coated with an oxide coating or with cadmium or zinc plate. Unless otherwise specified (see 6.1), all components, except cracks and forcing screws, shall have one of the following coatings:

3.14.2.1 Chromium plate. The finish shall be electrodeposited metals consisting of nickel followed by chromium, the minimum thickness being 0.0002-inch nickel and 0.00001-inch chromium.

3.14.2.2 Cadmium plate. The coating shall be electrodeposited cadmium not less than 0.0003 inch thick and shall be subjected to a chemical or electrochemical chromating.

3.14.2.3 Zinc plate. The coating shall be electrodeposited zinc not less than 0.0005 inch thick and shall be subjected to a chemical or electrochemical chromating.

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3.14.2.4 Hot phosphating. The coating shall consist of a heavy chemically produced phosphate followed with a coating of rust preventive oil.

3.14.2.5 Oxide coating. The coating shall consist of a heavy chemically produced oxide followed with a coating of rust preventive oil.

3.15 Workmanship. Workmanship shall be of the highest grade throughout and equal in every respect to commercial practice. Puller kits shall be free from rust, fins, burrs, external sharp edges, corners or surfaces, and defects which may impair their serviceability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table XXVII.

TABLE XXVII. Sampling data

Category	Sample Unit	Inspection Level	Acceptable Quality Level	AQL Expressed In Terms Of	Reference
Visual examination	One each	II	1.5 Major 2.5 Minor	Percent defective	4.3.1
Dimensional examination	One each	II	1.5	Percent defective	4.3.2
Testing	One each	S-4	2.5	Percent defective	4.4

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking. Defects are listed in table XXVIII.

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TABLE XXVIII. Classification of defects

Categories	Defects
Critical:	None defined.
Major:	
101	Type, class style, and size of puller kit not as specified.
102	Materials defective or not as specified, evidence of cracks, deep pits, voids, forging flash or heavy scale.
103	Screw threads stripped, torn, cracked, drunken (not concentric), or not perfect full threads.
104	Failure of components or assemblies to operate properly.
105	Improper fit between mating components.
106	Component(s) bent or otherwise malformed.
107	Component(s) missing.
108	Component(s) not within the specified dimensions.
109	Alike components not interchangeable.
110	Carrying case fails to accommodate the puller and components specified.
Minor:	
201	Tool not smoothly finished ; evidence of rust, fins, burrs, rough surfaces, or sharp edges.
202	Coating missing, incomplete, or not as specified; evidence of cracking, peeling, flaking, uncoated, or nonadherent areas.
203	Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanent.
204	Operating instructions missing, illegible, incorrect, or not securely attached to the carrying case (types I and II only).

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1, 4.4.1.1, and 4.4.2.

4.4.1 Load test. Tests of pulling strength shall be conducted to determine conformance with the loads specified in section 3. The test shall be conducted on a hydraulic press or by any other method approved by the inspector. The test load shall be applied through the forcing screw by supplying the torque load required to give a direct pull on the jaws and other assembled parts of the puller. The forcing screw thread shall be lubricated while applying load during test. With the minimum load applied, there shall be no buckling or deformation on any part of the pullers, or puller kits, nor reveal any permanent set, indicating that the elastic limit has been exceeded.

4.4.1.1 Pullers and puller kits. The minimum load specified for each type, class, style, and size shall be applied with the puller adjusted to maximum rated reach, and spread, with the forcing screw advanced one-third its length through the bridging member, except where these testing dimensions are shown in the respective tables.

4.4.2 Hardness and casehardness test. Hardness and casehardness test of draw nuts, expansion pins, adjusting screws, and collets for the type I, class 2 and 3 puller kits shall be accomplished in accordance with FED-STD-151. Method No. 243.1 shall be used for determining the Rockwell hardness, and Method No. 321.1 shall be used for determining depth of case hardening (see 3.9.2.1, 3.9.2.4, 3.9.2.5, 3.9.2.6, 3.9.3.4, 3.9.3.5, and 3.9.3.6).

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4.5 Examination of preparation for delivery. Inspection to determine compliance with preparation for delivery requirements shall be accomplished in accordance with MIL-H-15424 for puller.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. The mechanical puller kits shall be preserved, packaged, and packed in accordance with MIL-H-15424, as specified for puller, gear, and bushing. The level of preservation and packaging shall be A or C, and the level of packing A, B, or C, as specified (see 6.1).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. Marking shall be in accordance with MIL-H-15424.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, style, and size (see 1.2.1 and applicable table).
(c) Style of type II puller kit for which type V puller attachment is required (see 3.10.1.9).
(d) Size or sizes of pulling adapter blades and installing plates and whether adapter plates will be furnished for wet or dry removal or both (see 3.11.2).
(e) Sizes and number of nuts required for the type IV puller kit (see 3.12.7.1).
(f) If specific coating or coatings are qequired, state the king (s) required (see 3.14.2).
(g) Selection of applicable levels of preservation, packaging, and packing (see 5.1).

6.1.1 When repair parts are required, specify the following:

(a) The manufacturer's name or trademark of the puller kits for which the part is required.
(b) Quantity and proper name or description of part required.
(c) The type, class, style, and size of puller and puller kit for which the part is required.
(d) Any other data applicable to the puller or puller kit which will assist in identifying proper repair parts.

6.2 Superseding data. This specification supersedes all types, classes, and styles of puller kits in Federal Specification GGG-P-781; (types XIV, XV, XVI, XVII and XVIII have been renumbered I, II, III, IV and V respectively in this specification).

MILITARY INTEREST:

Custodians:

Army - GL
Air Force - 84

Review activities:

Army - AV, GL
Navy - SH
Air Force - 85

User activities:

Army - AT, ME, WC, CD, HD, HF
Navy - AS, YD, OS, CG, MG, GH, HD
Air Force - TG, TA, SG

MILITARY COORDINATION ACTIVITIES:

Air Force - 84

Preparing activity:

GSA-FSS

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